NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

DEPARTMENT

OF

COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

SUBJECT CODE: BE01R00041

SUBJECT NAME: MATHEMATICS-I

Subject Code:	**BE01R00041**	**Subject Name**	**Credit:**	4
Semester:	I	Mathematics-I	**Hours:**	120
Department:		Computer Science and Engineering		

MODULE: 1

BASIC CALCULUS

Evaluation of improper integrals of Type-I and Type-II, Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions

Topic 1: Evaluation of Improper Integral Type-I and II

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate the improper integral $\int_0^\infty \frac{1}{x^2}dx$ **[NLJIET]**	3	CO1	EL
2	Evaluate the improper integral $\int_1^\infty \frac{1}{x^2}dx$ **[NLJIET]**	3	CO1	EL
3	The following integral is an improper integral of which type? $\int_0^\infty \frac{dx}{x^2+1}$ **(Jan-2020) (Mar-2022) [NLJIET]**	3	CO1	RM
4	Evaluate $\int_1^\infty \frac{1}{1+x^2}dx$ **[NLJIET]**	3	CO1	EL
5	Define Improper Integrals of type-I. Evaluate $\int_{-1}^\infty \frac{dt}{t^2+5t+6}$ **(Jul-2023) [NLJIET]**	3	CO1	RM
6	Evaluate $\int_0^\infty \frac{dv}{(1+v^2)\ (1+\tan^{-1} v)}$ **(Jul-2024) [NLJIET]**	3	CO1	EL
7	Evaluate the improper integral $\int_5^\infty \frac{5x}{(1+x^2\ )^3}dx$ **[NLJIET]**	3	CO1	EL
8	Evaluate $\int_{-\infty}^\infty \frac{1}{e^x+e^{-x}}dx$ by using improper integral. **[NLJIET]**	3	CO1	EL
9	Evaluate improper integral $\int_0^3 \frac{1}{\sqrt{3-x}}dx$ **[NLJIET]**	3	CO1	EL
10	Evaluate $\int_0^3 \frac{dx}{x-1}$ if possible. **[NLJIET]**	3	CO1	EL
11	Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}}dx$ **[NLJIET]**	4	CO1	EL
12	Evaluate $\int_{-\infty}^\infty \frac{dx}{1+x^2}$ **[NLJIET]**	4	CO1	EL
13	Discuss Type I and Type II improper integrals with examples of each. Evaluate $\int_2^\infty \frac{x+3}{(x-1)(x^2+1)}dx$ **[NLJIET]**	4	CO1	RM
14	Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$ **[NLJIET]**	4	CO1	EL

Topic 2: Convergence and divergence of the integrals

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Discuss the convergence of the integral $\int_0^\infty \frac{1}{x^2}dx$ **[NLJIET]**	3	CO1	AP
2	Check the convergence of $\int_0^5 \frac{1}{x^2}dx$ **(Jul-2024) [NLJIET]**	3	CO1	EL
3	Discuss the convergence of the integral $\int_0^4 \frac{1}{x^{-2}}dx$ **[NLJIET]**	3	CO1	AP
4	Test the convergence of $\int_0^\infty \frac{1}{1+x^2}dx$ **[NLJIET]**	3	CO1	EL
5	Check the convergence of $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$ **[NLJIET]**	3	CO1	EL
6	Define Improper integral of both the kinds. Check the convergence of $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$. **[NLJIET]**	3	CO1	RM
7	Investigate the convergence of $\int_0^1 \frac{dx}{1-x}$ **(Jun-2025-New) [NLJIET]**	3	CO1	EL
8	Determine whether the integral $\int_0^3 \frac{dx}{x-1}$ converges or diverges. **(Jun-2019) [NLJIET]**	3	CO1	RM
9	Check the convergence of $\int_0^1 \frac{dx}{\sqrt{1-x}}$ **[NLJIET]**	3	CO1	EL
10	Test the convergence of $\int_0^\infty \frac{\cos x^2}{x^2}dx$ **[NLJIET]**	3	CO1	EL
11	Determine whether $\int_0^{\frac{\pi}{2}} \sec x dx$ converges or diverges. **[NLJIET]**	3	CO1	RM
12	Check the convergence of $\int_4^\infty \frac{3x+5}{x^4+7}dx$ **(Mar-2021) [NLJIET]**	3	CO1	EL
13	Does the improper integral $\int_0^\infty e^{-5x}dx$ converge or diverge? Justify. **[NLJIET]**	3	CO1	RM
14	Prove that $\int_1^\infty \frac{5}{e^x+3}dx$ is convergent. **[NLJIET]**	3	CO1	UN
15	Is $\int_4^\infty \frac{\sin^2 x}{\sqrt{x}\,(x-1)}dx$ convergent? **[NLJIET]**	4	CO1	RM

16	Discuss the convergence of the improper integral $\int_1^\infty \frac{\log x}{x^2}\, dx$**(Mar-2023)[NLJIET]**	4	CO1	AP
17	Prove that $\int_1^\infty \frac{\cos x}{x^2} dx$ converges. **[NLJIET]**	4	CO1	UN
18	Write the different forms of improper integrals. Check the convergence of an improper integral $\int_5^\infty \frac{7x+4}{x^2+9}\, dx$ using comparison test. **[NLJIET]**	4	CO1	EL
19	Investigate convergence of the following integrals:$\int_5^\infty \frac{5x}{(1+x^2)^3}\, dx$ **(Jan-2019) [NLJIET]**	4	CO1	EL
20	Discuss the convergence of $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx$.**(Jan-2025-New) [NLJIET]**	4	CO1	AP
21	State Direct comparison test for improper integrals and using it show that $\int_1^\infty e^{-x^2} dx$ is convergent. **[NLJIET]**	4	CO1	AP

Topic 3 : Beta and Gamma functions and their properties

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define the following: Beta function **[NLJIET]**	3	CO1	RM
2	Define Beta function. Find $\beta(5, 4)$. **[NLJIET]**	3	CO1	AP
3	Find $\beta\left(\frac{9}{2},\frac{7}{2}\right)$ **[NLJIET]**	3	CO1	AN
4	Find the value of $\beta(\frac{7}{2}, \frac{5}{2})$ **(Jan-2020) [NLJIET]**	3	CO1	AN
5	Find the value of $\beta(2,3)$ **[NLJIET]**	3	CO1	AN
6	Express$\int_0^1 x^m(1-x^n)^p dx$ in terms of Gamma function **[NLJIET]**	3	CO1	AP
7	Define Beta function. Evaluate $\int_0^1 x^4\left(1-\sqrt{x}\right)^5 dx$. **(Jul-2023) [NLJIET]**	3	CO1	AP
8	Define Beta and Gamma functions and Evaluate $\int_0^\infty \frac{x^3\,(1+x^2)}{(1+x)^{10}} dx$ **(Jan-2025-New) [NLJIET]**	3	CO1	AP

9	Define Gamma function and Evaluate $\int_0^\infty x^6 e^{-2x} dx$. **[NLJIET]**	**3**	**CO1**	**AP**
10	Show that $\Gamma(\mathrm{m}+1) = \mathrm{m}!$, where Γ is the Gamma function and m is a positive integer. **[NLJIET]**	**3**	**CO1**	**UN**
11	Prove that $Gamma(n) = (n-1)Gamma(n-1)$. **(Aug-2022) [NLJIET]**	**3**	**CO1**	**UN**
12	Prove that $\beta(m,n) = \beta(m+1,n) + \beta(m,n+1)$**[NLJIET]**	**3**	**CO1**	**UN**
13	Using the Beta and Gamma functions evaluate the integral $\int_{-1}^{1}(1-x^2)^n dx$, where n is a positive integer. **[NLJIET]**	**3**	**CO1**	**AP**
14	Define Beta and Gamma functions. What is the relationship between Beta and Gamma functions? **(Jan-2024) (Jun-2025-New) [NLJIET]**	**4**	**CO1**	**RM**
15	Define Beta function and Prove that (i) $\Gamma\frac{1}{2} = \sqrt{\pi}$ (ii)$\beta(m,n) = \beta(m,n+1) + \beta(m+1,n)$ **[NLJIET]**	**4**	**CO1**	**AP**
16	Evaluate $\int_3^7 \sqrt[4]{(x-3)(7-x)}dx$ using the concept of Gamma and Beta function **[NLJIET]**	**4**	**CO1**	**AP**
17	Define Gamma function and evaluate $\int_0^\infty e^{-x^2}\, dx$ **(Jun-2019) [NLJIET]**	**4**	**CO1**	**AP**
18	Investigate convergence of the following integrals:$\int_0^\infty \frac{x^{10}\,(1+x^5)}{(1+x)^{27}} dx$ **(Jan-2019) [NLJIET]**	**4**	**CO1**	**EL**
19	Prove that$\int_0^2 x^4(8-x^3)^{-1/3} dx = \frac{16}{3}\beta(\frac{5}{3},\frac{2}{3})$ **[NLJIET]**	**4**	**CO1**	**UN**
20	Define Beta function and evaluate $\int_0^1 x^3\left(1-\sqrt{x}\right)^5 dx$. **(Jul-2024) [NLJIET]**	**4**	**CO1**	**AP**
21	Evaluate $\int_0^\infty 3^{-4x^2} dx$.**(Jul-2024) [NLJIET]**	**4**	**CO1**	**EL**

Topic 4: Applications of definite integral

(Areas of Surfaces of Revolution)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define improper integral. Find the area between the curve	**4**	**CO1**	**AP**

	$y^2 = \frac{x^2}{1-x^2}$ and its asymptotes using improper integral **[NLJIET]**			
2	Find the surface area of the solid generated by revolving the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$ About its base. **[NLJIET]**	**4**	**CO1**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the area of the surface of revolution of a quadrant of a circular arc as obtained by revolving it about a tangent at one of its ends. **(Jul-2024) [NLJIET]**	**7**	**CO1**	**AN**
2	Find the surface area of a solid generated by the revolution of the circle $r = 2acos\theta$ about the initial line. **(Jan-2025-New) [NLJIET]**	**7**	**CO1**	**AN**

Topic 5: Applications of definite integral

(Volume using cross-sections)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Determine volume of a sphere of radius a by the solids of revolution. **[NLJIET]**	**3**	**CO1**	**RM**
2	Find the volume of the region between the curve $y = \sqrt{x}$, $0 \le x \le 4$ and the line $x = 4$ revolved about X –axis to generate a solid. **(Mar-2021) [NLJIET]**	**3**	**CO1**	**AN**
3	Find the volume of the solid that results when the region enclosed by the curves $y = x^2$ and $x = y^2$ is revolved about the Y-axis. **[NLJIET]**	**3**	**CO1**	**AN**
4	The graph of $y = x^2$ between $x = 1$ and $x = 2$ is rotated around the X −axis. Find the volume of a solid so generated. **[NLJIET]**	**3**	**CO1**	**AN**
5	The region bounded between the graph of $y = x^2$, and	**3**	**CO1**	**AN**

	$y = -x + 2$ and is rotating around the line $x = 4$.Find its volume. **[NLJIET]**			
6	Find the volume of solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 2$ *and* $x = 0$ about the line $y = 2$. **[NLJIET]**	**4**	**CO1**	**AN**
7	Define the volume of solid of revolution by disk and use it to find the volume of the solid that is obtained when the region under the curve $y = \sqrt{x}$ over the interval $[1, 4]$ is revolved about X -axis. **[NLJIET]**	**4**	**CO1**	**AP**
8	Find the volume of the region between the curve y $=\sqrt{x}$, $0 \leq x \leq 3$ and the line $x = 3$ revolved about the x- axis to generate a solid. **[NLJIET]**	**4**	**CO1**	**AN**
9	Find the volume of solid generated by revolving the region bounded by $y = x^2, y = 0$ *and* $x = 2$ about the $x - axis$. **(Jul-2023) [NLJIET]**	**4**	**CO1**	**AN**
10	Find the volume of solid generated by revolving the region between the parabola $x = y^2$ and the lines $x = 1$ about the line $x = 1$.**[NLJIET]**	**4**	**CO1**	**AN**
11	Find the volume of solid generated by revolving the region bounded by x= y^2 and the lines $x = 0, x = 2$ about the x-axis. **[NLJIET]**	**4**	**CO1**	**AN**
12	Use the shell method to find volume of solid generated by revolving the region by $y = x,\ x = 0$ *and* $y = 1$ about x-axis. **[NLJIET]**	**4**	**CO1**	**AP**
13	Find the volume of solid of revolution, obtained by rotating the area bounded below the line $2x + 3y = 6$ in the first quadrant, about the x–axis. **[NLJIET]**	**4**	**CO1**	**AN**
14	Use cylindrical shells to find volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$. **[NLJIET]**	**4**	**CO1**	**AP**

15	Find the volume of the solid obtained by rotating the region bounded by the curve $y = x^3$, $y = 8 \; and \; x = 0$ about the y-axis. **[NLJIET]**	**4**	**CO1**	**AN**
16	Using volume by slicing method, find the volume of a cylinder with radius 'r' and height 'h' .**[NLJIET]**	**4**	**CO1**	**AP**
17	Find the volume of a cone with height 4 cm and radius of base 4 cm. Use the method of slicing. **[NLJIET]**	**4**	**CO1**	**AN**
18	Find the volume of the pyramid whose base is a square with side L and whose height is h. **[NLJIET]**	**4**	**CO1**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the volume of solid generated by rotating the region bounded by $y = x^2 - 2x$ & $y = x$ about the lines $y = 4$. **(Mar-2022) [NLJIET]**	**7**	**CO1**	**AN**
2	The region between the curve $y = \sqrt{x}$, $0 \le x \le 4$ and the x-axis is revolved about the x-axis to generate a solid. Find its volume **(Jun-2025-New) [NLJIET]**	**7**	**CO1**	**AN**
3	The circle $x^2 + y^2 = a^2$ is rotated about x-axis to generate a sphere . Find its volume and sketch the region. **(Jan-2025-New) [NLJIET]**	**7**	**CO1**	**AN**
4	Using cylindrical shells, find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x}$ from 0 to 1. **[NLJIET]**	**7**	**CO1**	**AP**
5	Find the volume of the solid obtained by rotating the region enclosed by the curves $y = x \; and \; y = x^2$ about the x-axis. **(Jan-2019) [NLJIET]**	**7**	**CO1**	**AN**
6	Find the volume of the solid generated by rotating the plane region bounded by the curves $y = \frac{1}{x}, x = 1 \; and \; x = 3$ about the $X - axis$. **(Jan-2020) [NLJIET]**	**7**	**CO1**	**AN**

MODULE: 2

SINGLE-VARIABLE CALCULUS (DIFFERENTIATION)

Taylor's and Maclaurin's theorem for a function of one variable, Taylor's and Maclaurin's series of a function using statement of the theorems; Extreme values of functions; Indeterminate forms and L' Hospital's rule.

Topic 1: Taylor's series of a function using statement of the theorems

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Expand $log\ sin\ x$ in powers of $(x-2)$. **[NLJIET]**	3	CO2	AP
2	Find Taylor series generated by $f(x)=\frac{1}{x}$ at a = 2. Where, if any where does the series converge to $\frac{1}{x}$? **[NLJIET]**	3	CO2	AN
3	Find the Taylor series expansion of $f(x)=x^3-2x+4, a=2$ **[NLJIET]**	3	CO2	AN
4	Expand f (x) = $x^3 - 6x^2 + 14x + 1$ in Taylor's series about x = 2. **[NLJIET]**	3	CO2	AP
5	Expand $3x^3+8x^2+x-2$ in powers of $x-3$. **[NLJIET]**	3	CO2	AP
6	Express the polynomial $x^3 + 7x^2 + x - 6$ in powers of $(x - 1)$ **[NLJIET]**	3	CO2	AP
7	Express $5+4(x-1)^2-3(x-1)^3+(x-1)^4$ in ascending powers of *x*. **[NLJIET]**	3	CO2	AP
8	Express $(x-1)^4+2(x-1)^3+5(x-1)+2$ in ascending power of x. **[NLJIET]**	3	CO2	AP
9	Expand $\sin(x+\frac{\pi}{4})$ in powers of x by using the Taylor's series. Also, find the value of sin 46° **[NLJIET]**	3	CO2	AP
10	Use Taylor's series to estimate sin 38° **(Jun-2019) [NLJIET]**	3	CO2	AN
11	Using Taylor's expansion theorem, expand $f(x)=e^x$ at x=0. **[NLJIET]**	4	CO2	AN
12	Expand e^x in powers of $(x-1)$ by Taylor's series. **[NLJIET]**	4	CO2	AP

13	Find the Taylor series generated by $f(x) = e^x \; at \; x = 2$ **(Jun-2025-New) [NLJIET]**	**4**	**CO2**	**AN**
14	Expand $\log x$ in powers of $(x - 1)$ by Taylor theorem and hence find the value of $\log(1.1)$ **[NLJIET]**	**4**	**CO2**	**AP**
15	Expand $2x^3 + 7x^2 + x - 1$ in powers of $(x - 2)$. **(Mar-2022) [NLJIET]**	**4**	**CO2**	**AP**
16	Express $f(x) = 2x^3 + 3x^2 - 8x + 7$ in terms of $(x - 2)$. **(Jul-2024) [NLJIET]**	**4**	**CO2**	**AP**
17	Use Taylor series to represent function $f(x) = 2x^3 + x^2 + 3x - 8$ in powers of $(x - 1)$.**(Jul-2023) [NLJIET]**	**4**	**CO2**	**AN**
18	Expand the polynomial $f(x) = x^5 + 2x^4 - x^2 + x + 1$ in power of $x + 1$. **[NLJIET]**	**4**	**CO2**	**AP**
19	Expand the function $f(x) = x^4 - 11x^3 + 43x^2 - 60x + 14$ in powers of $(x - 3)$. **[NLJIET]**	**4**	**CO2**	**AP**
20	Expand $\sin(x + \frac{\pi}{4})$ in powers of x. Hence find the value of $\sin 44°$ **[NLJIET]**	**4**	**CO2**	**AP**
21	Find the Taylor's series expansion of $f(x) = \sin x$ in the power of $(x - \frac{\pi}{2})$. Hence obtain $\sin 91°$ **[NLJIET]**	**4**	**CO2**	**AN**
22	Find the Taylor series expansion of $f(x) = tan\, x$ in the powers of $(x - \frac{\pi}{4})$, showing at least four nonzero terms. Hence find the value of $\tan 46°$ **[NLJIET]**	**4**	**CO2**	**AN**
23	Find the expansion of $\tan\left(x + \frac{\pi}{4}\right)$ using Taylor series in ascending power of x up to x^4 and find $\tan 43°$ **[NLJIET]**	**4**	**CO2**	**AN**
24	Expand $\tan\left(\frac{\pi}{4} + x\right)$ in powers of x by using the Taylor's series. Also, find the value of $\tan 44^0$ **[NLJIET]**	**4**	**CO2**	**AP**
25	Expand $cos\left(\frac{\pi}{4} + x\right)$ in power of x. Hence, find the value of $cos 46^{\circ}$ **[NLJIET]**	**4**	**CO2**	**AP**
26	Express $\cos(a + h)$ as a series in powers of h and hence evaluate $\cos 44°$. **[NLJIET]**	**4**	**CO2**	**AP**

27	Assuming the validity of the expansion, show that, $\log x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots \text{for } 0 < x \le 2$, **[NLJIET] OR** Use Taylor's Series to find the expansion of $f(x) = \log_e x$ in power of $(x-1)$.**[NLJIET]**	**4**	**CO2**	**AN**
28	Using Taylor's series find $\sqrt[3]{27.12}$ correct to four decimal places. **[NLJIET]**	**4**	**CO2**	**AP**
29	Find the approximate value of $\sqrt{25.15}$ using Taylor's theorem. **(Mar-2023) [NLJIET]**	**4**	**CO2**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Expand $\sin(\frac{\pi}{4} + x)$ in power of x hence find the value of $\sin 45°$ **[NLJIET]**	**7**	**CO2**	**AP**
2	Expand $log\, cos(x + \frac{\pi}{4})$ using Taylor's theorem in ascending powers of x and hence find the value of $\log\cos(48°)$ correct up to three decimal places. **[NLJIET]**	**7**	**CO2**	**AP**
3	Find Taylor series expansion of the function $\sqrt{x+h}$ in powers of h and hence find the value of $\sqrt{18}$ correct up to three decimal places. **[NLJIET]**	**7**	**CO2**	**AN**
4	State Taylor's series for one variable and hence find $\sqrt{36.12}$ **[NLJIET]**	**7**	**CO2**	**AN**

Topic 2: Maclaurin's series of a function using statement of the theorems

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find Maclaurin series of the function $x^4 - 2x^3 - 5x + 4$ **[NLJIET]**	**3**	**CO2**	**AN**
2	Find the Maclaurin's series for the function $f(x) = e^x\ upto\ x^4$ **[NLJIET]**	**3**	**CO2**	**AN**

3	Expand $e^{x\sin x}$ in power of x up to the terms containing x^6. **(Jan-2019) [NLJIET]**	**3**	**CO2**	**AP**
4	Find the Maclaurin series of function $\tan x$ up to terms containing x^5. **[NLJIET]**	**3**	**CO2**	**AN**
5	Find the Maclaurin's series for $cosx$. **(Jan-2024) [NLJIET]**	**3**	**CO2**	**AN**
6	Expand $\cosh x$ using Maclaurin's series method. **[NLJIET]**	**3**	**CO2**	**AP**
7	Prove that $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$. **[NLJIET]**	**3**	**CO2**	**UN**
8	Prove that $\tan^{-1}\left(\frac{x(3-4x^2)}{\sqrt{1-x^2}\,(1-4x^2)}\right) = 3\left(x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots\right)$ **[NLJIET]**	**3**	**CO2**	**UN**
9	Use the binomial series and the fact that $\frac{d(\sin^{-1} x)}{dx} = (1-x^2)^{-\frac{1}{2}}$ to generate the first four nonzero terms of the Taylor series for $\sin^{-1} x$ and hence obtain the first five nonzero terms of the Taylor series for $\cos^{-1} x$. **[NLJIET]**	**3**	**CO2**	**AP**
10	Using Maclaurin's series prove that $\tanh^{-1} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$ **[NLJIET]**	**3**	**CO2**	**AP**
11	Prove that $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$ **(Jul-2024) [NLJIET]**	**3**	**CO2**	**UN**
12	Find Maclaurin's series for $f(x) = \sqrt{x}$ & $g(x) = x$ **[NLJIET]**	**3**	**CO2**	**AN**
13	Find the expansion of $\log(1+x)$ **[NLJIET]**	**3**	**CO2**	**AN**
14	Expand $\log(1+e^x)$ in ascending power of x as far as term containing x^4 **[NLJIET]**	**3**	**CO2**	**AP**
15	Expand $e^{\sin x}$ by Maclaurin's series up to the terms containing x^4 **[NLJIET]**	**4**	**CO2**	**AP**
16	Expand $f(x) = e^x \cos x$ in powers x of up to the terms containing x^4.**[NLJIET]**	**4**	**CO2**	**AP**
17	Expand $\frac{e^x}{\cos x}$ up to four terms by Maclaurin's series. **[NLJIET]**	**4**	**CO2**	**AP**
18	Expand Maclaurin series for $\sin x$ and $\cos x$ **[NLJIET]**	**4**	**CO2**	**AP**

19	Find the Maclaurin's series of $cosx$ and use it to find the series of sin^2x.**(Mar-2022) [NLJIET]**	**4**	**CO2**	**AN**
20	Find the Maclaurin series for the function $f(x) = \frac{1}{\sqrt{4-x}}$ **[NLJIET]**	**4**	**CO2**	**AN**
21	Find Maclaurin's series for $e^{2x}\sinh x$ and show at least up to x^4 term. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Expand $\tan^{-1} x$ up to the first four terms by Maclaurin's series and hence prove that $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)$ **[NLJIET]**	**7**	**CO2**	**AP**
2	Obtain the Maclaurin's series of $log_e(1+x)$ and hence find the series of $log_e\left(\frac{1+x}{1-x}\right)$ and then obtain approximate value of $log_e\left(\frac{11}{9}\right)$. **[NLJIET]**	**7**	**CO2**	**AN**
3	Find the Maclaurin series for $f(x) = (1+x)^k$where k is any real number.Using it ,find the Maclaurin series for the function $\frac{1}{1-x}$.**(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**AN**
4	Expand e^xinto the power series of x. Find number of terms required to calculate e with an error less than 10^{-4}. **(Jan-2025-New) [NLJIET]**	**7**	**CO2**	**AP**

Topic 3: Extreme values of functions

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the absolute maximum and absolute minimum value of the function $f(x) = x^2 - 1$ on the interval $-1 \le x \le 2$ **[NLJIET]**	**3**	**CO2**	**AN**
2	Find the interval on which the function $f(x) = x^4 - 4x^3 + 10$ is increasing and decreasing. **[NLJIET]**	**3**	**CO2**	**AN**

3	Find the value of local maximum and local minimum of the function $x^3 - 9x^2 + 24x + 2$. **[NLJIET]**	**3**	**CO2**	**AN**
4	Find the value of local maximum and local minimum of the function $x^4 - 4x^3 + 4x^2$. **[NLJIET]**	**4**	**CO2**	**AN**
5	Find the points of local extrema for $f(x) = sin2x - x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$,Also find their values. **[NLJIET]**	**4**	**CO2**	**AN**
6	Write second derivative test for local extrema and Find the extrema of $f(x) = x^4 - 4x^3 + 10$. **(Jan-2025-New) [NLJIET]**	**4**	**CO2**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the local extreme value of function $f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$**(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**AN**

Topic 4: Indeterminate forms and L' Hospital's rule.

$\left(\frac{0}{0}\right)$ Form

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\lim_{x \to \frac{1}{2}} \frac{cos^2 \pi x}{e^{2x} - 2ex}$ **[NLJIET]**	**3**	**CO2**	**EL**
2	Use L'Hospital rule to find $\lim_{x \to \frac{\pi}{2}} \frac{\log \sin x}{(\pi - 2x)^2}$ **[NLJIET]**	**3**	**CO2**	**AP**
3	Evaluate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$ **[NLJIET]**	**3**	**CO2**	**EL**
4	Evaluate $\lim_{x \to \frac{\pi}{2}} \frac{2x - \pi}{\cos x}$ **[NLJIET]**	**3**	**CO2**	**EL**
5	Evaluate using L'Hospital's rule $\lim_{x \to 0} \left(\frac{x - \sin x}{x^3}\right)$ **[NLJIET]**	**3**	**CO2**	**EL**
6	Evaluate using L'Hospital's rule $\lim_{x \to 0} \left(\frac{a^x - b^x}{x}\right)$ **[NLJIET]**	**3**	**CO2**	**EL**
7	Evaluate $\lim_{x \to 0} \frac{\tan x - x}{x^2 \tan x}$ **[NLJIET]**	**3**	**CO2**	**EL**
8	Evaluate $\lim_{x \to 0} \frac{x - tanx}{x^3}$ **[NLJIET]**	**3**	**CO2**	**EL**

9	Evaluate $\lim_{x\to 0}\frac{e^x+e^{-x}-x^2-2}{sin^2x-x^2}$ **[NLJIET]**	**3**	**CO2**	**EL**
10	Using L'Hospital's rule evaluate $\lim_{x\to 0}\frac{tanx-sinx}{x^2}$ **[NLJIET]**	**3**	**CO2**	**AP**
11	Evaluate using L' Hospital rule $\lim_{x\to 0}\frac{x^2+2\cos x-2}{x\sin^3 x}$ **[NLJIET]**	**3**	**CO2**	**EL**
12	Evaluate $\lim_{x\to 0}\frac{1-cosx}{sinx}$ **[NLJIET]**	**3**	**CO2**	**EL**
13	Evaluate $\lim_{x\to 0}\frac{1-cosx}{x^2}$ **[NLJIET]**	**3**	**CO2**	**EL**
14	Evaluate $\lim_{x\to 0}\frac{3x-\sin x}{x}$, using L'Hospital's rule **(Jun-2025-New) [NLJIET]**	**3**	**CO2**	**EL**
15	Find $\lim_{x\to a}\frac{\sin x-\sin a}{(x-a)^2}$ **(Aug-2022) [NLJIET]**	**3**	**CO2**	**AN**
16	Evaluate: $\lim_{x\to 0}\frac{e^x-e^{-x}-2x}{x-sinx}$**(Jul-2023) [NLJIET]**	**3**	**CO2**	**EL**
17	Using L'Hospital's rule,Evaluate $\lim_{x\to 1}\frac{x-x^x}{1+\log x-x}$ **.(Jul-2024) [NLJIET]**	**3**	**CO2**	**EL**
18	Evaluate$\lim_{x\to 0}\frac{2x-xcosx-sinx}{x^3}$ **[NLJIET]**	**4**	**CO2**	**EL**
19	Evaluate $\lim_{x\to 0}\frac{xe^x-\log(1+x)}{x^2}$ **(Jan-2020) [NLJIET]**	**4**	**CO2**	**EL**
20	Evaluate $\lim_{x\to 0}\frac{(1+x)^{\frac{1}{x}}-e+\frac{1}{2}ex}{x^2}$ **[NLJIET]**	**4**	**CO2**	**EL**
21	Evaluate the limits: (i) $\lim_{x\to 0}\frac{x(\cos x-1)}{\sin x-x}$; (ii)$\lim_{x\to 0}\frac{2\sqrt{1+x}-2-x}{2\,sin^2x}$ **[NLJIET]**	**4**	**CO2**	**EL**
22	Evaluate $\lim_{x\to 0}\frac{tan^2x-x^2}{x^2tan^2x}$ **(Mar-2021) (Mar-2023) [NLJIET]**	**4**	**CO2**	**EL**
23	Evaluate $\lim_{x\to 0}\frac{e^x-e^{-x}-2\log(1+x)}{x\,sinx}$ **(Mar-2022) [NLJIET]**	**4**	**CO2**	**EL**
	$\left(\frac{\infty}{\infty}\right)$ **Form**			
1	Evaluate using L'Hospital rule $\lim_{x\to a}\frac{\log(x-a)}{\log(e^x-e^a)}$ **[NLJIET]**	**3**	**CO2**	**EL**
2	Evaluate $\lim_{x\to a}\frac{log(e^x-e^a)}{\log(x-a)}$ **[NLJIET]**	**3**	**CO2**	**EL**
3	Evaluate $\lim_{n\to\infty}\frac{\ln n}{n}$ **[NLJIET]**	**3**	**CO2**	**EL**

4	Use L'Hospital rule, Evaluate $\lim_{x\to 0}\frac{\log x}{\cot x}$ **[NLJIET]**	**3**	**CO2**	**EL**
5	Evaluate $\lim_{x\to 0}\frac{\log(\sin x)}{(\cot x)}$ **[NLJIET]**	**3**	**CO2**	**EL**
	$(0\times\infty)$ **Form**			
1	Use L'Hospital rule to evaluate $\lim_{x\to 1}(1-x)\tan\left(\frac{\pi x}{2}\right)$**[NLJIET]**	**3**	**CO2**	**EL**
2	Evaluate $\lim_{x\to\infty}(a^{\frac{1}{x}}-1)\,x$ **[NLJIET]**	**3**	**CO2**	**EL**
3	Using L'Hospital Rule, Evaluate $\lim_{x\to 0}\frac{1}{x}(1-xcotx)$ **[NLJIET]**	**3**	**CO2**	**EL**
4	Evaluate $\lim_{x\to\infty} x\tan\frac{1}{x}$ **[NLJIET]**	**3**	**CO2**	**EL**
5	Evaluate $\lim_{x\to\frac{\pi}{2}} cosx\ logtanx$ **[NLJIET]**	**3**	**CO2**	**EL**
6	Evaluate $\lim_{x\to 0^+} x\ln x$ **(Jan-2024) [NLJIET]**	**3**	**CO2**	**EL**
	$(\infty-\infty)$ **Form**			
1	Evaluate $\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{x}\right)$ **[NLJIET]**	**3**	**CO2**	**EL**
2	State L'Hospital's Rule. Use it to evaluate $\lim_{x\to 0}\left[\frac{1}{x^2}-\frac{1}{\sin^2 x}\right]$ **(Jan-2019)[NLJIET]** Find $\lim_{x\to 0}\left[\frac{1}{x^2}-\frac{1}{\sin^2 x}\right]$ **(Jan-2025-New) [NLJIET]**	**3**	**CO2**	**EL**
3	Evaluate using L' Hospital rule $\lim_{x\to 1}\left[\frac{1}{\log x}-\frac{x}{x-1}\right]$ **[NLJIET]**	**3**	**CO2**	**EL**
4	Use L' Hospital's rule to to find $\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$ **(Jun-2019) [NLJIET]**	**3**	**CO2**	**AN**
5	Evaluate $\lim_{x\to\frac{\pi}{2}}(secx-tanx)$ **[NLJIET]**	**3**	**CO2**	**EL**
6	Evaluate: $\lim_{x\to 0}(cosecx-\cot x)$ **[NLJIET]**	**3**	**CO2**	**EL**
7	Evaluate $\lim_{x\to 2}\left[\frac{1}{x-2}-\frac{1}{\log(x-1)}\right]$ **[NLJIET]**	**3**	**CO2**	**EL**
	(0^0) **Form**			
1	Evaluate $\lim_{x\to\frac{\pi}{2}}(\cos x)^{\frac{\pi}{2}-x}$ **[NLJIET]**	**3**	**CO2**	**EL**
2	Evaluate $\lim_{x\to 0}\sin x^{\tan x}$ **[NLJIET]**	**3**	**CO2**	**EL**

3	Evaluate the limit: $\lim_{x\to1}(1-x^2)^{\frac{1}{\log(1-x)}}$ **[NLJIET]**	**3**	**CO2**	**EL**
	(1^∞) **Form**			
1	Evaluate $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\tan x}$ **[NLJIET]**	**3**	**CO2**	**EL**
2	Evaluate $\lim_{x\to0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}}$ **[NLJIET]**	**3**	**CO2**	**EL**
3	Evaluate $\lim_{x\to0}(a^x+x)^{\frac{1}{x}}$ **[NLJIET]**	**3**	**CO2**	**EL**
4	Evaluate $\lim_{x\to0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{3x}}$ **[NLJIET]**	**3**	**CO2**	**EL**
5	Evaluate $\lim_{x\to0}\left(\frac{1^x+2^x+3^x}{3}\right)^{\frac{1}{x}}$ **[NLJIET]**	**3**	**CO2**	**EL**
6	Evaluate $\lim_{x\to0}(\cos x)^{\cot x}$ **[NLJIET]**	**3**	**CO2**	**EL**
7	Evaluate $\lim_{x\to1}(2-x)^{\tan\frac{\pi x}{2}}$ **[NLJIET]**	**3**	**CO2**	**EL**
8	Evaluate $\lim_{x\to\frac{\pi}{2}}(1-\cos x)^{\tan x}$ **[NLJIET]**	**3**	**CO2**	**EL**
9	Find $\lim_{x\to0}\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)^{\frac{1}{x}}$ **[NLJIET]**	**4**	**CO2**	**EL**
10	Evaluate $\lim_{x\to0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{x}}$ **[NLJIET]**	**4**	**CO2**	**EL**
	(∞^0) **Form**			
1	Evaluate $\lim_{x\to0}\left(\frac{1}{x}\right)^{\tan x}$ **[NLJIET]**	**3**	**CO2**	**EL**
2	Using L'Hospital Rule, evaluate $\lim_{x\to0}(\cot x)^{\sin x}$ **[NLJIET]**	**3**	**CO2**	**EL**
3	Evaluate: $\lim_{x\to0}\left(\frac{1}{x}\right)^{1-\cos x}$ **[NLJIET]**	**4**	**CO2**	**EL**

MODULE: 3
SEQUENCES AND SERIES

Sequence of numbers and its convergence, Infinite series; Tests for convergence (Telescoping series, Geometric series test, Integral test, p test, comparison test, D' Alembert's ratio test, Cauchy's root test), Alternating series test; Power series, Radius and interval of convergence, Conditional and Absolute convergence of a power series .

Topic 1: Sequence of numbers and its convergence

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Show that the sequence $\{u_n\}$ whose n[th] term is $u_n = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots \frac{1}{n!}\ n \in N$ is monotonic increasing and bounded. Is it convergent? **[NLJIET]**	3	CO3	UN
2	Show that the sequence $\frac{n}{n^2+1}$ is monotonic decreasing and bounded. Is it convergent? **[NLJIET]**	3	CO3	UN
3	Show that the sequence $\left\{\frac{3}{n+3}\right\}$ is a decreasing sequence. **[NLJIET]**	3	CO3	UN
4	Define monotonic sequence. Is the sequence $\left\{\frac{1}{n^2}\right\}$ monotonic? **(Jan-2024) [NLJIET]**	3	CO3	RM
5	Find convergence of sequence $\left\{\frac{(-1)^n}{2}\right\}$ **[NLJIET]**	3	CO3	AN
6	Test the convergence of the sequence $\{2-(-1)^n a\}$ **[NLJIET]**	3	CO3	EL
7	Test the convergence of sequence $\left(\frac{n^2+n}{2n^2-n}\right)$. **[NLJIET]**	3	CO3	EL
8	Using Sandwich theorem find limit of sequence $(a_n) = \left(\frac{\cos n}{n}\right)$ **(Jul-2023) [NLJIET]** State Sandwich theorem for sequence and discuss the convergence of the sequence $\left\{\frac{\cos n}{n}\right\}$ **(Jan-2025-New) [NLJIET]**	3	CO3	AP
9	State Sandwich theorem and using it find $\lim_{x\to 0} g(x)$ if $3 - x^2 \le g(x) \le 3\sec x$,for all x. **[NLJIET]**	3	CO3	AP
10	Show that the sequence $\{u_n\}$,where $u_n = \frac{\sin n}{n}$ converges to zero. **(Jul-2024) [NLJIET]**	3	CO3	UN

11	Define the convergence of a sequence (a_n) and verify whether the sequence whose n^{th} term is $a_n = \left(\frac{n+1}{n-1}\right)^n$ convergent or not. **(Jun-2019) [NLJIET]**	4	CO3	RM

Topic 2: Tests for convergence

Telescoping Series

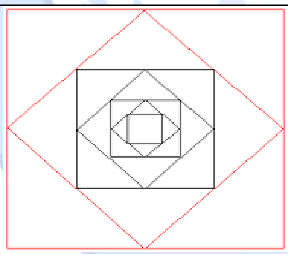
SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Discuss the convergence of the series and find its sum $\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$ **(Jan-2025-New) [NLJIET]** Find the sum of the series $\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$ **(Jun-2025-New) [NLJIET]**	4	CO3	EL
2	Determine whether the following series converges or diverges. Find the sum of the series $\sum_{n=1}^{\infty}[\tan^{-1}(n) - \tan^{-1}(n+1)]$,if converges. **[NLJIET]**	4	CO3	RM
3	Test the convergence of the series $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{1}{n^2+n+1}\right)$ and if it is convergent then find its sum. **[NLJIET]**	4	CO3	EL
4	Show that series $\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$ is convergent. Hence find sum of the series. **(Jul-2023) [NLJIET]**	4	CO3	AP

Topic 3: Geometric Series

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Determine whether the following series converge or diverge. Find the sum of the series $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$ if converges. **[NLJIET]**	3	CO3	AP
2	Find the sum of following series if it converges; $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$ **[NLJIET]**	3	CO3	AN
3	Test the convergence of $\sum_{n=0}^{\infty}\frac{3^{2n}}{2^{3n}}$ **[NLJIET]**	3	CO3	EL

4	Find the sum of the series $\sum_{n\geq 2}\frac{1}{4^n}$ **(Jan-2019) [NLJIET]**	3	CO3	AN
5	Show that the series $\sum_{n=0}^{\infty}\left(\frac{e}{\pi}\right)^n$ is convergent and find its sum. **(Mar-2023) [NLJIET]**	3	CO3	AN
6	Let S=$\sum_{n=1}^{\infty} n\alpha^n$ where $\lvert\alpha\rvert<1$. Find the value of α in (0,1) such that S=2α. **[NLJIET]**	3	CO3	AN
7	Find the value of *b* for which $1 + e^b + e^{2b} + e^{3b} + \ldots + e^{nb} + \ldots = 9$ **[NLJIET]**	3	CO3	AN
8	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty}\frac{4^n+5^n}{6^n}$ **(Jan-2019) [NLJIET]**	3	CO3	EL
9	Test the convergence of the series $\sum_{n=0}^{\infty}\left(\frac{1}{2^n}+\frac{1}{3^n}\right)$ **[NLJIET]**	3	CO3	EL
10	Find the sum of the series $\sum_{n=0}^{\infty}\frac{2^n-1}{3^n}$ **(Jul-2023) [NLJIET]**	3	CO3	AN
11	You drop a ball from a meter above a flat surface. Each time the ball hits the surface after falling a distance h, it rebounds a distance *rh*, where *0 < r < 1*. Find the total distance ball travels up and down where $a = 6\ m$ and r = $\frac{2}{3}$ m. **[NLJIET]**	3	CO3	AN
12	 The figure shows that the first seven of a sequence of squares. The outermost square has an area of 4 m^2. Each of the other squares is obtained by joining the midpoints of the sides of the squares before it. Find the sum of the areas of all squares in the infinite sequence. **[NLJIET]**	3	CO3	AN
13	Define geometric series and show that the geometric series is (i) convergent if $\lvert r\rvert<1$ (ii) divergent if $r \geq 1$ **[NLJIET]**	4	CO3	RM
14	Prove that $1+\frac{2}{3}+\frac{4}{9}+\frac{8}{27}+\frac{16}{81}+\cdots$ converges and find it's sum.	4	CO3	AN

	[NLJIET]			
15	Test the convergence of the series. If converges find the sum.$\sum_{n=0}^{\infty}\frac{1+2^n}{5^n}$ [NLJIET]	4	CO3	EL
16	Define the Geometric series and find the sum of the following series $\sum_{n=1}^{\infty}\frac{3^{n-1}-1}{6^{n-1}}$ [NLJIET]	4	CO3	AN
17	Test the convergence of the series $\sum_{n=0}^{\infty}\frac{4^n+3}{5^n}$, Also find the sum of series if it is convergent. **(Jan-2025-New) [NLJIET]**	4	CO3	EL

Topic 4: Cauchy's Integral test, The P-series

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence of $\sum_{n=1}^{\infty}\frac{1}{n^{3/2}}$ **[NLJIET]**	3	CO3	EL
2	Check whether the series $\sum\frac{1}{\sqrt{n}}$ is Convergent or Divergent [NLJIET]	3	CO3	EL
3	State the Integral Test and determine the convergence of $\sum_{n=1}^{\infty}\frac{2\tan^{-1}n}{1+n^2}$ [NLJIET]	3	CO3	RM
4	Test the convergence of the series $\sum_{n=2}^{\infty}\frac{1}{nlogn}$ [NLJIET]	3	CO3	EL
5	Use integral test to show that the following infinite series is convergent $\sum_{n=1}^{\infty}\frac{1}{n(1+log^2 n)}$ **(Mar-2022) [NLJIET]**	3	CO3	UN
6	Does $\sum_{n=1}^{\infty}e^{-n}$ series converge or diverge? **[NLJIET]**	4	CO3	EL
7	Test the convergence of $\sum_{n=3}^{\infty}\frac{1}{n\log n\sqrt{log^2 n-1}}$ **[NLJIET]**	4	CO3	EL
8	Write the statement of Cauchy's integral test. Test the convergence of the following series $\sum_{n=2}^{\infty}\frac{1}{n(logn)^a}$, $for\ 0 \le a \le 1.$ **[NLJIET]**	4	CO3	EL
9	Test the convergence of the series $\sum ne^{-n^2}$ **(Jan-2024)** [NLJIET]	4	CO3	EL
10	Is $\sum_{n=1}^{\infty}\frac{1}{n^p}$ convergent for p > 1? Justify your answer.	4	CO3	EL

	(Aug-2022) [NLJIET]			

Topic 5: The Comparison Test, The Limit Comparison Test

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	For which values of p do the series $\sum_{n=1}^{\infty} \frac{n^p}{\sqrt{n+1}+\sqrt{n}}$ is convergent. **[NLJIET]**	3	CO3	AN
2	For which values of p do the series $\frac{2}{1^p}+\frac{3}{2^p}+\frac{4}{3^p}+\cdots$ is convergent. **[NLJIET]**	3	CO3	AN
3	Discuss the convergence of $\sum_{n=1}^{\infty}[\sqrt{n^2+1}-n]$ **[NLJIET]**	3	CO3	EL
4	Test for the convergence the series; $\sum_{n=1}^{\infty}[\sqrt[3]{n^3+1}-n]$ **[NLJIET]**	3	CO3	EL
5	Test the convergence of the series $\sum_{n=1}^{\infty}[\sqrt{n^4+1}-\sqrt{n^4-1}]$ **[NLJIET]**	3	CO3	EL
6	Test the convergence of the series $\sum_{n=1}^{\infty}\sqrt{n+1}-\sqrt{n}$ **(Mar-2021) [NLJIET]**	3	CO3	EL
7	Using comparison test, discuss the convergence of $\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{1}{n}\right)$ **[NLJIET]**	3	CO3	EL
8	Examine the convergence of $\sum_{n=1}^{\infty}\sin\left(\frac{1}{n}\right)$ **[NLJIET]**	3	CO3	EL
9	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{2n+1}{n^2+2n+1}$. **(Mar-2023) [NLJIET]**	3	CO3	EL
10	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{n+1}{n^3+3}$ **[NLJIET]**	3	CO3	EL
11	Determine whether series converges or diverges $\sum_{n=1}^{\infty}\frac{1}{n^2+n+1}$. **[NLJIET]**	3	CO3	EL
12	Test the series $\sum_{n=1}^{\infty}\frac{1}{n^2+1}$ for convergence or divergence. **(Jun-2025-New) [NLJIET]**	3	CO3	EL

13	Check for convergence/divergence $\sum_{n=1}^{\infty} \frac{5n^3-3n}{n^2(n-2)(n^2+5)}$ **[NLJIET]**	3	CO3	EL
14	Determine whether the series $\sum_{n=1}^{\infty} \frac{(n-1)}{n^3+3}$ converges or diverges. **[NLJIET]**	3	CO3	EL
15	Examine the convergence of $\sum_{n=1}^{\infty} \frac{2n^2+3n}{5+n^5}$ **[NLJIET]**	3	CO3	EL
16	Determine convergence or divergence of series $\sum_{n=1}^{\infty} \frac{(2n^2-1)^{\frac{1}{3}}}{(3n^3+2n+5)^{\frac{1}{4}}}$ **[NLJIET]**	3	CO3	EL
17	Test the convergence of the series $\frac{1\cdot 2}{3^2\cdot 4^2} + \frac{3\cdot 4}{5^2\cdot 6^2} + \frac{5\cdot 6}{7^2\cdot 8^2} + \ldots$ **[NLJIET]**	3	CO3	EL
18	Test for convergence the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots$ **[NLJIET]**	3	CO3	EL
19	Test the convergence of $\frac{1}{1\cdot 2\cdot 3} + \frac{3}{2\cdot 3\cdot 4} + \frac{5}{3\cdot 4\cdot 5} + \cdots$**[NLJIET]**	3	CO3	EL
20	Test the convergence of $\frac{3}{1^2-3} + \frac{3}{2^2-3} + \frac{3}{3^2-3} + \frac{3}{4^2-3} + \cdots$ **[NLJIET]**	3	CO3	EL
21	Check for the convergence of the following: $\sum_{n=1}^{\infty} \frac{1}{1^2+2^2+\cdots+n^2}$ **[NLJIET]**	3	CO3	EL
22	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$. **(Jul-2024) [NLJIET]**	4	CO3	EL
23	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$ **(Jan-2020) [NLJIET]**	4	CO3	EL
24	State the *p*-series test. Test the convergence of series $\sum_{n=0}^{\infty} \frac{n+1}{n^3-3n+2}$ **(Jan-2019) [NLJIET]**	4	CO3	EL
25	Find the sum of the series$\sum_{n\geq 1} \frac{4}{(4n-3)(4n+1)}$ **(Jun-2019) [NLJIET]**	4	CO3	AN

Topic 6: Ratio Test

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{n^3+2}{2^n+2}$ **[NLJIET]**	3	CO3	EL
2	Use Ratio test to check the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n+1}{3^n+1}$ **(Mar-2022) [NLJIET]**	3	CO3	AP
3	Test the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ **[NLJIET]**	3	CO3	EL
4	Investigate the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{7^n}$. **(Aug-2022) [NLJIET]**	3	CO3	EL
5	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{n2^n(n+1)!}{3^n n!}$ **[NLJIET]**	3	CO3	EL
6	Using ratio test discuss convergence of the series $\sum_{n=1}^{\infty} \frac{(n+3)!}{3!n!3^n}$ **(Jul-2023) [NLJIET]**	3	CO3	AP
7	Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ **[NLJIET]**	3	CO3	EL
8	Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$. **(Jun-2025-New) [NLJIET]**	3	CO3	EL
9	Test the convergence of $\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$ by Ratio Test. **[NLJIET]**	3	CO3	EL
10	Investigate the Convergence of the series $\sum_{n=1}^{\infty} \frac{4^n n! n!}{(2n)!}$ **(Jan-2025-New) [NLJIET]**	3	CO3	EL
11	Test the convergence of the following series $1+\frac{x}{2}+\frac{x^2}{3^2}+\frac{x^3}{4^3}+\cdots, x>0$ **[NLJIET]**	3	CO3	EL
12	Test for convergence the series $2+\frac{3}{2}x+\frac{4}{3}x^2+\frac{5}{4}x^3+\cdots$ **[NLJIET]**	3	CO3	EL
13	Test the convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n^3+1}$ **(Mar-2021) [NLJIET]**	4	CO3	EL
14	Test the convergence of series $\sum_{n=0}^{\infty} \frac{4^n+1}{5^n}$ **[NLJIET]**	4	CO3	EL
15	Check the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n+5}{3^n}$ **(Mar-2021) [NLJIET]**	4	CO3	EL

16	Investigate the convergence of $\sum_{n=0}^{\infty} \frac{2^n+5}{3^n}$ **(Jan-2024) [NLJIET]**	4	CO3	EL
17	Investigate the convergence of $\sum_{n=1}^{\infty} \frac{2^n(n!)^2}{(2n)!}$. **(Aug-2022) [NLJIET]**	4	CO3	EL
18	Test the D'Alembert's ratio test. Discuss the convergence of the following series $\sum_{n=1}^{\infty} \frac{4^n(n+1)!}{n^{n+1}}$ **(Jan-2019) [NLJIET]**	4	CO3	EL
19	Test the convergence of the series $1 + \frac{2^2}{2!} + \frac{3^2}{3!} + \frac{4^2}{4!} +$..**[NLJIET]**	4	CO3	EL
20	Test for the convergence for the series $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots$ **[NLJIET]**	4	CO3	EL
21	Test the convergence of the following series $\sqrt{\frac{1}{2}}x + \sqrt{\frac{2}{5}}x^2 + \sqrt{\frac{3}{10}}x^3 + \cdots, x > 0$ **[NLJIET]**	4	CO3	EL
22	Find value of ***x*** for which the given series $\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \cdots$ converges. **[NLJIET]**	4	CO3	AP
23	Test the convergence of $x + \frac{x^2}{2^3} + \frac{x^3}{3^3} + \cdots$ **[NLJIET]**	4	CO3	EL

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \cdots$, where $x > 0$ **(Jan-2020) [NLJIET]**	7	CO3	EL
2	Test for the convergence the series; $\frac{1}{1\cdot2\cdot3} + \frac{x}{4\cdot5\cdot6} + \frac{x^2}{7\cdot8\cdot9} + \cdots$ **(Jan-2019)[NLJIET] OR** Determine all the positive values of x for which the series $\frac{1}{1\cdot2\cdot3} + \frac{x}{4\cdot5\cdot6} + \frac{x^2}{7\cdot8\cdot9} + \cdots$ converges. **(Mar-2023) [NLJIET]**	7	CO3	EL

Topic 7: Cauchy's Root Test

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level

1	Test the convergence of the following series $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$ **[NLJIET]**	3	CO3	EL
2	Determine whether series converges or diverges $\sum_{n=1}^{\infty}\left(\frac{2n+3}{3n+12}\right)^{n}$ **[NLJIET]**	3	CO3	EL
3	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{n}{e^{-n}}$ **[NLJIET]**	3	CO3	EL
4	Test for convergence the series whose nth term is $\left(1+\frac{1}{\sqrt{n}}\right)^{-n^{\frac{3}{2}}}$ **[NLJIET]**	3	CO3	EL
5	Test the convergence or divergence of the series $\sum_{n=1}^{\infty}\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}$ **[NLJIET]**	3	CO3	EL
6	Test the convergence of the series $\frac{1}{3}+\left(\frac{2}{5}\right)^2+\left(\frac{3}{7}\right)^3+\cdots+\left(\frac{n}{2n+1}\right)^n+\cdots$ **(Jan-2020) [NLJIET]**	3	CO3	EL
7	Test Cauchy's root test. Discuss the convergence of the series $\sum_{n=2}^{\infty}\frac{n}{(\log n)^n}$ **(Jan-2019) [NLJIET]**	4	CO3	EL
8	Test the convergence of the series $\sum_{n=1}^{\infty}\left(\frac{1}{1+n}\right)^{n}$ **(Jan-2024) [NLJIET]**	4	CO3	EL
9	Write the statement of Cauchy's root test, for which value of x does the series, $\sum_{n=1}^{\infty}\left(\frac{n+1}{n+2}\right)^{n}x^n\ ; x>0$ is convergent. What can we say at the point x = 1? **[NLJIET]**	4	CO3	RM
10	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{n^n\,x^n}{(n+1)^n}\ ; x>0$ **[NLJIET]**	4	CO3	EL
11	Test the convergence of $\sum_{n=1}^{\infty}\frac{[(n+1)x]^n}{n^{n+1}}$ **[NLJIET]**	4	CO3	EL

Topic 8: Alternating Series Test (Leibnitz's Test), Absolute And Conditional convergence

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty} \frac{(-1)^n (n+1)^n}{(2n)^n}$ **[NLJIET]**	3	CO3	EL
2	Discuss the convergence of the following series $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$ **[NLJIET]**	3	CO3	EL
3	Discuss the convergence of the following series$\sum \frac{(-1)^{n-1}}{n\sqrt{n}}$ **[NLJIET]**	3	CO3	EL
4	Discuss the convergence of $1-\frac{1}{2\sqrt{2}}+\frac{1}{3\sqrt{3}}-\frac{1}{4\sqrt{4}}+\cdots$ **[NLJIET]**	3	CO3	EL
5	Test the convergence of the series $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots$ **(Jan-2020) [NLJIET]**	3	CO3	EL
6	Check for convergence of the series $\frac{1}{1\cdot 2}-\frac{1}{3\cdot 4}+\frac{1}{5\cdot 6}-\frac{1}{7\cdot 8}+\cdots$ **[NLJIET]**	3	CO3	EL
7	Discuss the convergence of the series $4-9+5+4-9+5+4-9+5 \ldots .\infty$ **[NLJIET]**	3	CO3	EL
8	Check the convergence of the series $\sum_{n=0}^{\infty}(-1)^n(\sqrt{n+\sqrt{n}}$ - $\sqrt{n}\,)$ **(Jun-2019) [NLJIET]**	4	CO3	EL
9	Show that $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ converges. **(Aug-2022) [NLJIET]**	4	CO3	UN
10	Show that the series $\frac{1}{1^p}-\frac{1}{2^p}+\frac{1}{3^p}-\frac{1}{4^p}+\cdots$ converges for p > 0 **[NLJIET]**	4	CO3	UN
11	Test the convergence of the series $\frac{1}{2}-\frac{2}{5}+\frac{3}{10}-\frac{4}{17}+\cdots$ **[NLJIET]**	4	CO3	EL
12	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2}$ **[NLJIET]**	4	CO3	EL
13	Test the convergence of the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$. **(Jun-2025-New) [NLJIET]**	4	CO3	EL

14	Is the series $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}\ldots$ convergent? Does it converge absolutely? Give reasons. **[NLJIET]**	4	CO3	EL
15	Check for absolute/conditional convergence of $\sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt{n}+\sqrt{n+1}}$ **[NLJIET]**	4	CO3	EL
16	Check the absolute and conditional convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt[3]{n}}$. **[NLJIET]**	4	CO3	EL
17	Determine absolute or conditional convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^n n^2}{n^3+1}$ **[NLJIET]**	4	CO3	EL
18	Is the series absolutely convergent or conditionally convergent? $1-\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{4}}+\cdots\infty$ **[NLJIET]**	4	CO3	EL
19	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{(n+1)sin\left\{\frac{(2n-1)\pi}{2}\right\}}{nlog(n+1)}$ **(Mar-2023) [NLJIET]**	4	CO3	EL

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^2}$**(Jan-2024)** **[NLJIET]**	7	CO3	EL
2	Define absolutely convergent series and conditionally convergent series. Investigate the convergence of the series $\sum_{n=1}^{\infty}\frac{\sin n}{n^2}$ **(Jun-2025-New) [NLJIET]**	7	CO3	EL

Topic 9: Power Series, Radius and interval of Convergence

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Test the convergence and Divergence of the following series $\sum_{n=1}^{\infty}n!\,(x-4)^n$ **[NLJIET]**	3	CO3	EL

2	For $\sum_{n=0}^{\infty}\frac{(2x+3)^{2n+1}}{n!}$ (a) find the series' radius of convergence. For what values of x does the series converges (b) absolutely, (c) conditionally? **[NLJIET]**	**3**	**CO3**	**AN**
3	For $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}\,(3x-1)^{n}}{n^{2}}$ (a) find the series' radius of convergence. For what values of x does the series converges (b) absolutely, (c) conditionally? **[NLJIET]**	**3**	**CO3**	**AN**
4	Find the interval of convergence of the series $1 + 2x + 3x^2 + 4x^3 \ldots$ **(Jan-2019) [NLJIET]**	**3**	**CO3**	**AN**
5	Test for convergence the series $x - \frac{x^3}{3} + \frac{x^5}{5} + \cdots, x > 0$ **[NLJIET]**	**3**	**CO3**	**EL**
6	Find the radius of convergence and interval of convergence of the series $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ **[NLJIET]**	**4**	**CO3**	**AN**
7	Determine the interval of convergence for the series $\sum_{n=1}^{\infty}\frac{x^n}{2n}$; $x > 0$ **[NLJIET]**	**4**	**CO3**	**AN**
8	For the series $\sum_{n=1}^{\infty}\frac{(-1)^n(x+2)^n}{n}$ find the series' radius and interval of convergence. For what values of $\boldsymbol{x}$ does the series converge absolutely, conditionally? **[NLJIET]**	**4**	**CO3**	**AN**
9	For what values of x do the power series $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}x^{2n-1}}{2n-1}$ converge? **[NLJIET] OR** For the series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1}$, find the radius and interval of convergence. **[NLJIET]**	**4**	**CO3**	**AN**
10	Find the radius of convergence and interval of convergence for the series $\sum_{n=0}^{\infty}\frac{(-3)^n\,x^n}{\sqrt{n+1}}$ **[NLJIET] OR** Find the interval of convergence of the series $\sum_{n=0}^{\infty}\frac{(-3)^n\,x^n}{\sqrt{n+1}}$ **[NLJIET]**	**4**	**CO3**	**AN**

11	Define Radius of convergence of the power series. Find it for power series $\sum_{n=1}^{\infty}\frac{(x-1)^n}{n^3 3^n}$ **(Jul-2023) [NLJIET]**	**4**	**CO3**	**RM**
12	Find the radius and interval of convergence of the power series $\sum_{n=0}^{\infty}\frac{(-5)^n x^n}{n!}$ **[NLJIET]**	**4**	**CO3**	**AN**
13	Find the radius of convergence and interval of convergence of the series $1-\frac{1}{2}(x-2)+\frac{1}{2^2}(x-2)^2+\cdots+\left(-\frac{1}{2}\right)^n(x-2)^n+\ldots$ **[NLJIET]**	**4**	**CO3**	**AN**
14	Test the convergence of the series $\frac{x}{2.5}-\frac{x^2}{2^2.10}+\frac{x^3}{2^3.15}-\frac{x^4}{2^4.20}+\cdots$ Find its radius of convergence. **[NLJIET]**	**4**	**CO3**	**EL**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the series radius and interval of convergence for $\sum_{n=0}^{\infty}\frac{(3x-2)^n}{n}$. For what values of x does the series converge absolutely? **(Jun-2019) [NLJIET]**	**7**	**CO3**	**AN**
2	Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=0}^{\infty}\frac{(-1)^n x^n}{3^n(n+1)}$ **(Jan-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
3	Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=1}^{\infty}\frac{(x-5)^n}{n^2}$ **(Jan-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
4	For what values of x does the series $\sum_{n=1}^{\infty}\frac{(x-3)^n}{n}$ converges? **(Jun-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
5	Find the interval of convergence for the series $\sum_{n=1}^{\infty}\frac{n(x+2)^n}{3^{n+1}}$ **[NLJIET]**	**7**	**CO3**	**AN**
6	Determine the radius and interval of convergence of the following infinite series	**7**	**CO3**	**AN**

	$x-\frac{x^2}{4}+\frac{x^3}{9}-\frac{x^4}{16}+\cdots$ **(Mar-2022) [NLJIET]**			
7	Find the interval of convergence of the series $x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots$ is convergent. **(Jan-2024) [NLJIET]**	**7**	**CO3**	**AN**
8	For which value of x does the series $\frac{x^2}{2}-\frac{x^3}{3}+\frac{x^4}{4}-\frac{x^5}{5}+\cdots$ is absolute or conditionally convergent or divergent? What is the radius of convergence of $\frac{x^2}{2}-\frac{x^3}{3}+\frac{x^4}{4}-\frac{x^5}{5}+\cdots$? **[NLJIET]**	**7**	**CO3**	**AN**

MODULE: 4

MULTIVARIABLE CALCULUS (DIFFERENTIATION)

Limit, Continuity and Differentiation for function of two or more variables, total derivative, gradient, directional derivatives; Tangent plane and Normal line to the surface $f(x,y,z)=c$; Extreme values for function of two variables (Maxima, minima and saddle points); Method of Lagrange multipliers.

Topic 1: Limit

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Determine whether $\lim_{(x,y)\to(0,0)}\frac{x^2y}{x^4+y^2}$, exist or not? If they exist find the value of the limit. **[NLJIET]**	**3**	**CO4**	**AN**
2	If $f(x,y)=\frac{xy^2}{x^2+y^4}$ does $\lim_{(x,y)\to(0,0)} f(x,y)$ exists? **(Jun-2025-New) [NLJIET]**	**3**	**CO4**	**AN**
3	Determine whether $\lim_{(x,y)\to(0,0)}\frac{2x^2y}{x^3+y^3}$, exist and find it if exists. **(Jan-2024) [NLJIET]**	**3**	**CO4**	**AN**
4	Determine whether $\lim_{(x,y)\to(0,0)}\frac{x^2y^2}{x^4+3y^4}$ exists or not? It they exist, find the value. **[NLJIET]**	**3**	**CO4**	**AN**
5	Show that the function $f(x,y)=\frac{2x^2y^2}{x^4+y^4}$ has no limit as (x, y) approaches to (0,0).**(Jun-2019) [NLJIET] OR**	**3**	**CO4**	**UN**

	Show that following limit does not exist using different path approach $\lim_{(x,y)\to(0,0)} \frac{2x^2y^2}{x^4+y^4}$ **(Mar-2022) [NLJIET]**			
6	Determine whether $\lim_{(x,y)\to(\mathbf{0},\mathbf{0})} \frac{x^6-y^2}{x^3y}$ exist, find it if exists. **(Jan-2025-New) [NLJIET]**	**3**	**CO4**	**AN**
7	Find $\lim_{(x,y)\to(0,0)} \frac{xy}{y^2-x^2}$ **[NLJIET]**	**3**	**CO4**	**AN**
8	Evaluate $\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$ **[NLJIET]**	**3**	**CO4**	**EL**
9	Show that $\lim_{(x,y)\to(\mathbf{0},\mathbf{0})} \frac{x^4-y^2}{x^4+y^2}$ does not exists. **(Jul-2023) [NLJIET]**	**4**	**CO4**	**UN**
10	Does limit of function $\frac{2xy}{x^2+3y^2}$ exists $at\ x = 0, y = 0$? **[NLJIET]**	**4**	**CO4**	**EL**

Topic 2: Continuity

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Show that $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 \quad , & (x,y) = (0,0) \end{cases}$ is continuous at every point except at the origin. **[NLJIET]**	**3**	**CO4**	**UN**
2	Discuss the continuity of the given function $f(x,y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2} & ; x \neq 0, y \neq 0 \\ 0 & ; x = 0, y = 0 \end{cases}$ **(Jan-2019) [NLJIET]**	**3**	**CO4**	**UN**
3	If $f(x,y) = \frac{y-x}{y+x}$ and f(0,0) = 0, discuss the continuity of *f(x,y)* at (0,0)**[NLJIET]**	**3**	**CO4**	**UN**
4	Show that $f(x,y) = \begin{cases} \frac{x^3y}{x^4+y^4}, & (x,y) \neq (0,0) \\ 0 \quad , & (x,y) = (0,0) \end{cases}$ is not continuous at the (0,0). **(Mar-2023) [NLJIET]**	**3**	**CO4**	**UN**
5	Show that $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 \quad , & (x,y) = (0,0) \end{cases}$ is continuous everywhere except at origin. **[NLJIET] OR**	**3**	**CO4**	**UN**

	Discuss the continuity of the function $f(x,y)=\begin{cases}\frac{xy}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0 \quad , & (x,y)=(0,0)\end{cases}$ **[NLJIET]**			
6	Show that $f(x,y)=\begin{cases}\frac{2x^2y}{x^3+y^3}, & (x,y)\neq(0,0)\\ 0 \quad , & (x,y)=(0,0)\end{cases}$ is not continuous at the origin. **(Mar-2021) [NLJIET]**	**3**	**CO4**	**UN**
7	Determine the continuity of the function $f(x,y)=\begin{cases}(x^2+y^2)\,sin\left(\frac{1}{x^2+y^2}\right), & (x,y)\neq(0,0)\\ 0 & ,(x,y)=(0,0)\end{cases}$ at origin **[NLJIET]**	**3**	**CO4**	**AN**
8	Show that $f(x,y)=x^2+2y$ is continuous at $(1,2)$. **[NLJIET]**	**3**	**CO4**	**UN**
9	Show that the function $f(x,y)=\frac{x^2-y^2}{x^2+y^2}$ has no limit as (x,y) approaches to $(0,0)$ **[NLJIET] OR** Discuss continuity of $f(x,y)=\begin{cases}\frac{x^2-y^2}{x^2+y^2}, & for\ (x,y)\neq(0,0)\\ 0 \quad , & for\ (x,y)=(0,0)\end{cases}\ in\ R\times R$ **[NLJIET]**	**4**	**CO4**	**UN**
10	Show that $f(x,y)=\begin{cases}\frac{xy}{\sqrt{x^2+y^2}}, & (x,y)\neq(0,0)\\ 0 \quad , & (x,y)=(0,0)\end{cases}$ is continuous at the origin. **[NLJIET]**	**4**	**CO4**	**UN**
11	Discuss the continuity of $f(x,y)=\begin{cases}\frac{x}{\sqrt{x^2+y^2}}, & x\neq 0, y\neq 0\\ 2 \quad , & x=0, y=0\end{cases}$ at the origin. **[NLJIET]**	**4**	**CO4**	**UN**
12	Determine the set of points at which the given function is continuous $f(x,y)=\begin{cases}\frac{3x^2y}{x^2+y^2}, & if\ (x,y)\neq(0,0)\\ 0 \quad , & if\ (x,y)=(0,0)\end{cases}$ **[NLJIET]**	**4**	**CO4**	**AN**
13	Discuss the continuity of $f(x,y)=\begin{cases}\frac{x^2y^2}{x^4+4y^4}, & (x,y)\neq(0,0)\\ \frac{1}{5} \quad , & (x,y)=(0,0)\end{cases}$	**4**	**CO4**	**UN**

	[NLJIET]			

Topic 3: Differentiation for function of two or more variables

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	If $u = \frac{e^{x+y+z}}{e^x+e^y+e^z}$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 2u$. **[NLJIET]**	3	CO4	UN
2	If $u = log(tanx + tany)$ then prove that $sin2x\frac{\partial u}{\partial x} + sin2y\frac{\partial u}{\partial y} = 2u$. **[NLJIET]**	3	CO4	UN
3	If $x = rcos\theta$ and $y = rsin\theta$, find $\frac{\partial x}{\partial r}$ and $\frac{\partial r}{\partial x}$ **[NLJIET]**	3	CO4	AN
4	If resistors of R_1, R_2 and R_3 ohms are connected in parallel to make an R -ohm resistor, the value of R can be found from the equation $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. Find the value of $\frac{\partial R}{\partial R_2}$ when $R_1 = 30$, $R_2 = 45$ and $R_3 = 90$ ohms. **[NLJIET]**	3	CO4	AN
5	Verify that the function $u = e^{-\alpha^2k^2t}.sinkx$ is a solution of the heat conduction equation $u_t = \alpha^2 u_{xx}$. **(Jun-2019) [NLJIET]**	3	CO4	EL
6	Find all first and second order partial derivatives for $f(x,y) = x^2siny + y^2cosx$ Hence, verify mixed derivative Theorem. **[NLJIET]**	3	CO4	AN
7	If $z = x + y^x$, prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ **[NLJIET]**	3	CO4	UN
8	If $u = tan^{-1}\left(\frac{y}{x}\right)$ Provethat $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ **(Jan-2025-New) [NLJIET]**	3	CO4	UN
9	Show that $\frac{\partial^2 \Omega}{\partial x \partial y} = \frac{\partial^2 \Omega}{\partial y \partial x}$ where $\Omega = y + x^y$. **[NLJIET]**	3	CO4	UN
10	Let $w = f(x, y, z)$ be a function of three independent variables, write the formal definition of the partial derivative for $\frac{\partial f}{\partial z}$ at (x_0, y_0, z_0) . Using this definition find $\frac{\partial f}{\partial z}$ at (1,2,3) for $f(x,y,z) = x^2yz^2$. **[NLJIET]**	4	CO4	AN
11	If $f(x,y) = x^2y + xy^2$then find $f_x(1,2)$ and $f_y(1,2)$ by definition. **[NLJIET]**	4	CO4	AN

12	If $f(x) = x^3 + x^2y^3 - 2y^2$, find $f_x(2,1)$ and $f_y(2,1)$ **[NLJIET]**	**4**	**CO4**	**AN**
13	If $u = log(tanx + tany + tanz)$ them prove that $sin2x\frac{\partial u}{\partial x} + sin2y\frac{\partial u}{\partial y} + sin2z\frac{\partial u}{\partial z} = 2u$. **(Mar-2022)** **[NLJIET]**	**4**	**CO4**	**UN**
14	If $u = x^2y + y^2z + z^2x$ then find out $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ **(Jan-2019) [NLJIET]**	**4**	**CO4**	**AN**
15	The amount of work done by the heart's main pumping chamber, the left ventricle, is given by the equation $W = PV + \frac{V\delta v^2}{2g}$,where $\boldsymbol{W}$ is the work per unit time, $\boldsymbol{P}$ is the average blood pressure, $\boldsymbol{V}$ is the volume of blood pumped out during the unit of time, δ is the weight density of the blood, $\boldsymbol{v}$ is the average velocity of the existing blood, and $\boldsymbol{g}$ is the acceleration of the gravity which is constant. Find partial derivative of $\boldsymbol{W}$ with respect to each variable. **[NLJIET]**	**4**	**CO4**	**AN**
16	Determine whether $u(x,y) = ln\sqrt{x^2 + y^2}$ is a solution Laplace's equation. **[NLJIET]**	**4**	**CO4**	**AN**
17	If $u = x^3y + e^{xy^2}$ then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ **[NLJIET]**	**4**	**CO4**	**UN**
18	If $u = f(x + at) + g(x - at)$, Prove that $\frac{\partial^2 u}{\partial t^2} = a^2\frac{\partial^2 u}{\partial x^2}$. **(Mar-2021) [NLJIET]**	**4**	**CO4**	**UN**
19	Compute the four second order partial derivatives of $f(x,y) = xy^2 + 3x^2e^y$ **[NLJIET]**	**4**	**CO4**	**AN**
20	If $u = (x^2 + y^2 + z^2)^{\frac{m}{2}}$ then find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$. **[NLJIET]**	**4**	**CO4**	**AN**
21	If $u = x^2y + y^2z + z^2x$, find the values of (i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ (ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ **[NLJIET]**	**4**	**CO4**	**AN**

22	If $u = e^{3xyz}$ show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2)e^{3xyz}$ **[NLJIET]**	**4**	**CO4**	**UN**
23	If $u = e^{xyz}$ show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$ **[NLJIET]**	**4**	**CO4**	**UN**
24	For $u = tan^{-1}\left(\frac{y}{x}\right)$; Show that $(a)\ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ $(b)\ \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ **[NLJIET]**	**4**	**CO4**	**UN**
25	If $u = log(x^2 + y^2)$, verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ **[NLJIET]**	**4**	**CO4**	**EL**
26	If $u = log(x^2 + y^2)$, Show that that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ **(Jul-2023) [NLJIET]**	**4**	**CO4**	**UN**
27	If $u = log(x^3 + y^3 - x^2y - xy^2)$ prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 u = -\frac{4}{(x+y)^2}$. **[NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	If $\theta = t^n e^{-\frac{r^2}{4t}}$ then find n so that $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$ **(Jul-2024) [NLJIET]**	**7**	**CO4**	**AN**
2	If $u = x^2y + y^2z + z^2x$, prove that (i) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 = 6(x + y + z)$ (ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2(x + y + z)$ **[NLJIET]**	**7**	**CO4**	**UN**
3	If $u = ln(x^3 + y^3 + z^3 - 3xyz)$ then prove that (a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$ (b) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$ **[NLJIET]**	**7**	**CO4**	**UN**
4	If $u = x^2y + y^2z + z^2x$ then prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = 3(u_{xx} + u_{yy} + u_{zz})$ **[NLJIET]**	**7**	**CO4**	**UN**

Topic 3:Total Derivative

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find $\frac{dw}{dt}$ if $w = xy + z$, $x = \cos t$, $y = \sin t$, $z = t$. **[NLJIET]**	3	CO4	AN
2	If $z = e^{xy}, x = t\cos t, y = t\sin t$, find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$ **[NLJIET]**	3	CO4	AN
3	If $u = x^2y^3, x = logt, y = e^t$ find $\frac{du}{dt}$ **[NLJIET]**	3	CO4	AN
4	If $u = x^2 + y^2, x = a\cos t, y = b\sin t$ then find $\frac{du}{dt}$ **[NLJIET]**	3	CO4	AN
5	Find $\frac{dw}{dt}$ at $t = 0$ if $w = xy^2 + z^2$, where $x = cost, y = \sin t, z = t$. **[NLJIET]**	3	CO4	AN
6	If $z = xy^2 + x^2y, x = at^2, y = 2at, find \frac{dz}{dt}$ **[NLJIET]**	3	CO4	AN
7	If $u = sin^{-1}(x - y), x = 3t, y = 4t^3$ then show that $\frac{du}{dt} = \frac{3}{\sqrt{1-t^2}}$. **[NLJIET]**	3	CO4	UN
8	If $u = sin^{-1}(x - y), x = 3t$ and $y = 4t$ then find $\frac{du}{dt}$. **[NLJIET]**	3	CO4	AN
9	Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s. If $w = x + 2y + z^2$, $x = r/s, y = r^2 + log(s), z = 2r$. **[NLJIET]**	3	CO4	AP
10	If $w = x^2 + y^2, x = r - s, y = r + s$, using chain rule , prove that $\frac{\partial w}{\partial s} = 4s$. **(Jan-2024) [NLJIET]**	3	CO4	UN
11	If $u = x^2 - y^2$, where $x = 2r - 3s + 4, y = -r + 8s - 5$, Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$ **[NLJIET]**	3	CO4	AN
12	If $u = f(x,y)$ show that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = 0$ **[NLJIET]**	3	CO4	UN
13	Suppose f is a differentiable function of x and y and $g(u,v) = f(e^u + sinv, e^u + cosv)$. Use the following table to calculate $g_u(0,0), g_v(0,0), g_u(1,2)\ and\ g_v(1,2)$	4	CO4	AP

	f	g	f_x	f_y
(0,0)	3	6	4	8
(1,2)	6	3	2	5

(Jun-2019) [NLJIET]

14	Applying Chain rule find $\frac{dw}{dt}$ if $w = x^2y - y^2$, where $x = sint\ and\ y = e^t$ also find $\frac{dw}{dt}$ at $t = 0$ **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**AN**
15	If $z = x^2y + 3xy^4$, where $x = sin2t$ and $y = cost$ find $\frac{dz}{dt}\ when\ t = 0$ **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**AN**
16	If $z = tan^{-1}\left(\frac{x}{y}\right), x = u\ cosv, y = u\ sinv$ then find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ using the chain rule.**(Mar-2023) [NLJIET]**	**4**	**CO4**	**AN**
17	If $u = f(r)$ where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$ **[NLJIET] OR** If $u = f(r)$ where $x = r\cos\theta, y = r\sin\theta$ Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r)$ **[NLJIET]**	**4**	**CO4**	**UN**
18	If $u = x^4y + y^2z^3$ where $x = rse^t, y = rs^2e^{-t}$, and $z = r^2s\sin t$, find the value of $\frac{\partial u}{\partial s}$,When $r = 1, s = 1, t = 0$. **[NLJIET]**	**4**	**CO4**	**AN**
19	If $u = f(x - y, y - z, z - x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ **(Jan-2020) (Mar-2021) (Mar-2022) [NLJIET]**	**4**	**CO4**	**UN**
20	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$ **OR [NLJIET]** If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}, t = \frac{z}{x}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$. **[NLJIET]**	**4**	**CO4**	**UN**
21	If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$ **[NLJIET]**	**4**	**CO4**	**UN**

22	If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that $(y^2 - zx)\frac{\partial u}{\partial x} + (x^2 - yz)\frac{\partial u}{\partial y} + (z^2 - xy)\frac{\partial u}{\partial z} = 0$ **[NLJIET]**	4	CO4	UN
23	If $u = f(l, m)$ where $l = x + y$ and $m = x - y$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2\frac{\partial u}{\partial l}$ **[NLJIET]**	4	CO4	UN
24	If $u = f(x, y)$, where $x = e^s cost\ and\ y = e^s sint$, show that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s}\left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2\right]$ **[NLJIET]**	4	CO4	UN
25	If $z = f(x, y), x = e^u + e^{-v}, y = e^{-u} - e^v$ then prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$ **[NLJIET]**	4	CO4	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	If $u = f(r)$ and $r^2 = x^2 + y^2 + z^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r}f'(r)$ **[NLJIET]**	7	CO4	UN
2	If $u = r^m$ then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = m(m+1)r^{m-2}$ **[NLJIET] OR** If $r^2 = x^2 + y^2 + z^2$ and $v = r^m$, prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = m(m+1)r^{m-2}$. **[NLJIET] OR** If $u = (x^2 + y^2 + z^2)^m$ then find $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ **[NLJIET]**	7	CO4	UN
3	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, Prove that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$ **[NLJIET]**	7	CO4	UN

Topic 4: Jacobian

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find Jacobian $\frac{\partial(x,y)}{\partial(r,\theta)}$ for $x = rcos\theta\ , y = rsin\theta$ **(Jul-2023)** **[NLJIET]**	3	CO4	AN

2	If $x = rsin\theta cos\varphi$, $y = rsin\theta sin\varphi$, $z = rcos\theta$,then find the Jacobian $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}$ **(Mar-2022) [NLJIET]**	4	CO4	AN

Topic 5: Gradient & Directional Derivative

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	If $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\widehat{k}$ Prove that $\nabla^2 r^n = n(n+1)r^{n-2}$ **[NLJIET]**	3	CO4	UN
2	If $\bar{r} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$, show that $\nabla logr = \frac{1}{r}\hat{r}$ .Where $\hat{r}$ is unit vector. **[NLJIET]**	3	CO4	UN
3	Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$ **[NLJIET]**	3	CO4	UN
4	If $\phi = 3x^2y - y^3z^2$, find $grad\phi$ at the point (1,-2,-1). **[NLJIET]**	3	CO4	AN
5	Define gradient of a function. Use it to find directional derivative of $f(x,y,z) = x^3 - xy^2 - z$ at $P(1,1,0)$ in the direction of $\bar{a} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$ **(Jan-2019) [NLJIET]**	3	CO4	AN
6	Find the directional derivatives of $f = xy^2 + yz^2$ at the point $(2,-1,1)$ in the direction of vector $i + 2j + 2k$ **(Jan-2020) [NLJIET]**	3	CO4	AN
7	The temperature at any point in space is given by $T = xy + yz + zx$. Determine the derivative of T in the direction of the vector $3\hat{\imath} - 4\hat{k}$ at the point (1,1,1). **[NLJIET]**	3	CO4	AN
8	Find the derivative of $f(x,y) = x^2 + xy + y^2$ in the direction $\hat{\imath} + \hat{\jmath}$ at $P(1,1)$. **(Aug-2022) [NLJIET]**	3	CO4	AN
9	Find the directional derivative of $f(x,y,z) = xyz$ at the point $P(-1,1,3)$ in the direction the vector $\bar{a} = \hat{\imath} - 2\hat{\jmath} + 2\hat{k}$ **(Mar-2022) [NLJIET]**	3	CO4	AN

10	Find the directional derivative of $f(x,y) = x^2 sin2y$ at the point $(1,\frac{\pi}{2})$ in the direction of $v = 3\hat{\imath} - 4\hat{\jmath}$ **(Mar-2023) [NLJIET]**	**3**	**CO4**	**AN**
11	Find directional derivative of the function $f(x,y,z) = x^2 + 3y^2 + z^2$ at the point $P(1,2,3)$ in the direction of the vector $\hat{\imath} - 2\hat{k}$. **[NLJIET]**	**3**	**CO4**	**AN**
12	Find the directional derivative of the divergence of $\bar{F}(x,y,z) = xy\hat{\imath} + xy^2\hat{\jmath} + z^2\hat{k}$ at the point (2, 1, 2) in the direction of the outer normal to the sphere $x^2 + y^2 + z^2 = 9$ **[NLJIET]**	**3**	**CO4**	**AN**
13	Find the directional derivative of $4xz^2 + x^2yz$ at $(1,-2,-1)$ in the direction of $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$. **(Jul-2024) [NLJIET]**	**3**	**CO4**	**AN**
14	Find the directional derivative of the scalar point function $f = 3e^{2x-y+z}$ at the point $A(1,1,-1)$ in the direction towards the point $B(-3,5,6)$ **[NLJIET]**	**3**	**CO4**	**AN**
15	Find the directional derivative $D_u f(x,y)$ if $f(x,y) = x^3 - 3xy + 4y^2$ and u is the unit vector given by angle $\theta = \frac{\pi}{6}$. What is $D_u f(1,2)$? **(Jun-2019) [NLJIET]**	**3**	**CO4**	**AN**
16	Find the gradient of $f(x,y,z) = 2z^3 - 3(x^2 + y^2)z + \tan^{-1}(xz)$ at (1,1,1). **[NLJIET]**	**4**	**CO4**	**AN**
17	Find the directional derivative of the function $f(x,y) = x^2y^3 - 4y$ at the point $(2,-1)$ in the direction of the vector $v = 2\hat{\imath} + 5\hat{\jmath}$. **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**AN**
18	Find the directional derivative of the function $f(x,y,z) = xy^2 + yz^3$ at the point (2,-1,1) in the direction of the vector $\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$. **[NLJIET]**	**4**	**CO4**	**AN**
19	Find the directional derivative of $x^2y^2z^2$ at $(1,1,-1)$ along a direction equally inclined with coordinate axes. **[NLJIET]**	**4**	**CO4**	**AN**
20	Find the directional derivative of $\emptyset = 4xz^3 - 3x^2y^2z$ at the point (2,-1,2) in the direction (2,3,6). **[NLJIET]**	**4**	**CO4**	**AN**

21	Find the directional derivative of $\emptyset = 4xz^3 - 3x^2yz^2$ at the point $(2,-1,2)$ in the direction $2\bar{i} + 3\bar{j} + 6\bar{k}$ **[NLJIET]**	**4**	**CO4**	**AN**
22	Find the directional derivative of $f(x,y,z) = 2x^2 + 3y^2 + z^2$ at the point $(2,1,3)$ in the direction of the vector $i - 2k$. **(Jan-2024) [NLJIET]**	**4**	**CO4**	**AN**
23	Find the directional derivative of $\emptyset = x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of the line PQ where Q is the point $(5,0,4)$. **[NLJIET]**	**4**	**CO4**	**AN**
24	Find directional derivative of function $\emptyset = zx^2 + 2xy^2 + yz^2$ at point $(1,2,-1)$ in the direction of the vector $\overline{a} = 2i + 3j - 4k$. **[NLJIET]**	**4**	**CO4**	**AN**
25	Find directional derivative of $xy + yz + zx$ in the direction of $3\hat{\imath} - 2\hat{\jmath} + \hat{k}$ at (1, 2, 3). **[NLJIET]**	**4**	**CO4**	**AN**
26	Find the directional derivative of $f(x,y,z) = x^2z + y^3z^2 - xyz$ at (1,1,1) in the direction of the vector (−1,0,3). **[NLJIET]**	**4**	**CO4**	**AN**
27	Let $f = \ln r$, where $\bar{r} = x\,\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $r = \lvert \bar{r} \rvert$. Find f . **(Aug-2022) [NLJIET]**	**4**	**CO4**	**AN**

Topic 6: Tangent plane and Normal line to the surface

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$, at the point (1,2,3) **[NLJIET]**	**3**	**CO4**	**AN**
2	Find the equation of the tangent plane and the normal line of the surface $x^2 + 2y^2 + 3z^2 - 12 = 0$ at the point $(1,2,-1)$ **(Mar-2021) [NLJIET]**	**3**	**CO4**	**AN**
3	Show that the surfaces $z = xy - 2$ and $x^2 + y^2 + z^2 = 3$ have the same tangent plane at (1,1,1). **[NLJIET]**	**3**	**CO4**	**UN**
4	Find the equations of tangent plane and normal line at a point (3,4,5) to the surface $x^2 + y^2 - 4z = 5$. **[NLJIET]**	**3**	**CO4**	**AN**

5	Find the equation of the tangent plane and normal line to the surface $\frac{x^2}{2} - \frac{y^2}{3} = z$ at the point (2, 3,-1). **[NLJIET]**	**3**	**CO4**	**AN**
6	Find the equation of the tangent plane and normal line to the surface $2xz^2 - 3xy - 4x = 7$at $(1,-1,2)$ **[NLJIET]**	**3**	**CO4**	**AN**
7	Find the equations of the tangent plane at the point $(-2,1,-3)$ to the Ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$**(Jun-2025-New) [NLJIET]**	**3**	**CO4**	**AN**
8	find the equation of tangent plane and normal line at point (1,2,3) to the surface $x^2 + 2y^2 - 3z = 0$ **[NLJIET]**	**3**	**CO4**	**AN**
9	Find the equation of tangent plane and normal line to the surface $cos\pi x - x^2y + e^{xz} + yz = 4$ at the point $P(0,1,2)$. **[NLJIET]**	**3**	**CO4**	**AN**
10	Find the equations of tangent plane and normal line of the surface $x^2 + y^2 - z^2 = 4$ at the point (1,2,1). **[NLJIET]**	**3**	**CO4**	**AN**
11	Find the equation of tangent plane and the normal line to the surface $2x^2 + y^2 + 2z = 3$ at $(2,1,-3)$. **[NLJIET]**	**4**	**CO4**	**AN**
12	Find the tangent plane and normal line of the surface $f(x,y,z) = x^2 + y^2 + z - 9 = 0$ at the point $P(1,2,4)$. **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**AN**
13	Find the equation of the tangent plane and normal line to the surface $\mathrm{x}^2 + \mathrm{y}^2 + \mathrm{z}^2 = 3$ at the point (1,1,1). **(Jan-2020) [NLJIET]**	**4**	**CO4**	**AN**
14	Find the equations of tangent plane and normal line at the point $(-2,1,-3)$ to the Ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$.**[NLJIET]**	**4**	**CO4**	**AN**
15	Find the equations of the tangent plane and normal line to the surface $z = 2x^2 + y^2$ at the point$(1,1,3)$**[NLJIET]**	**4**	**CO4**	**AN**
16	Find the tangent plane and normal line to the surface $x^2yz + 3y^2 = 2xz^2 - 8z$ at the point $p(1,2,-1)$. **(Mar-2023) [NLJIET]**	**4**	**CO4**	**AN**

17	Find the equation of tangent plane and normal line to the surface $xyz = 6$ at (1,2,3) **[NLJIET]**	4	CO4	AN
18	Find the equation of tangent plane and normal line at a point (3,4,5) to the surface $x^2 + y^2 - 4z - 5 = 0$. **[NLJIET]**	4	CO4	AN
19	Find the equations of tangent plane and normal line to $x^2 + y^2 + z^2 = 81$ at the point $(-1,-4,8)$ **[NLJIET]**	4	CO4	AN
20	Find the tangent plane of $z = e^x \cos y$ at $P(0,0,0)$. **(Aug-2022) [NLJIET]**	4	CO4	AN
21	Find the equations of tangent plane and normal line to surface $\frac{x^2}{2} - \frac{y^2}{3} - z = 0 \ at \ (2,3,-1)$**(Jul-2023) [NLJIET]**	4	CO4	AN
22	Find the equations of tangent plane to $z = 3x^2 - xy$at the point (1,2,1)**(Jan-2024) [NLJIET]**	4	CO4	AN

Topic 7: Extreme values for function of two variables (Maxima, minima and saddle points)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain second derivative test for local extreme values. **(Aug-2022) [NLJIET]**	3	CO4	UN
2	Find all the local maxima, local minima, and saddle points of the function $f(x, y) = x^2 - 2xy + 2y^2 - 2x + 2y + 1$ **[NLJIET]**	3	CO4	AN
3	Discuss the maxima and minima of the function $x^2 + y^2 + 6x + 12$ **[NLJIET]**	3	CO4	UN
4	For what values of the constant k does the second derivative test guarantee that $f(x, y) = x^2 + kxy + y^2$ will have a saddle point at (0,0)? A local minimum at (0,0)? **(Jun-2019) [NLJIET]**	4	CO4	AN
5	Find the local maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$ **[NLJIET]**	4	CO4	AN
6	Examine $f(x, y) = x^3 + y^3 - 3axy$ for maximum and minimum values. **[NLJIET] OR**	4	CO4	EL

	Determine the points where the function $f(x,y) = x^3 + y^3 - 3axy$ has a maxima or minima. **[NLJIET]**			
7	For $f(x,y) = x^3 + y^3 - 3xy$, find the maximum and minimum values**. (Mar-2021) [NLJIET] OR** Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3xy$.**(Mar-2023) [NLJIET]**	**4**	**CO4**	**AN**
8	Find the extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. **[NLJIET]**	**4**	**CO4**	**AN**
9	Find a point within a triangle such that the sum of the squares of its distances from the three vertices is a minimum **[NLJIET]**	**4**	**CO4**	**AN**
10	Find the extreme values of $sinx + siny + \sin(x+y)$; $\left(0 \le x, y \le \frac{\pi}{2}\right)$ **[NLJIET]**	**4**	**CO4**	**AN**
11	Find all local maxima, local minima and saddle point of $f(x,y) = x^2 + y^2 + 4x + 6y + 13$**[NLJIET]**	**4**	**CO4**	**AN**
12	Find the maximum and minimum values and saddle point, if any of $f(x,y) = x^2 + y^2 + 2x + 4y + 9$ **[NLJIET]**	**4**	**CO4**	**AN**
13	Find the points of maxima and minima of $(x,y) = x + y + \frac{1}{x} + \frac{1}{y}$,Also find the maximum and minimum values of the function. **[NLJIET]**	**4**	**CO4**	**AN**
14	Find the maximum and minimum values $f(x,y) = 2(x^2 + y^2) - x^4 + y^4$ **[NLJIET]**	**4**	**CO4**	**AN**
15	Discuss the Maxima and Minima of the function $3x^2 - y^2 + x^3$**(Jan-2020) [NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find maxima and minima of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$. **[NLJIET] OR** Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$	**7**	**CO4**	**AN**

	(Mar-2022) (Jul-2024) [NLJIET]			
2	Find all local maxima, local minima and saddle points of the function $f(x,y) = x^3 + 3xy^2 - 15x + y^3 - 15y$. **(Jul-2023) [NLJIET]**	**7**	**CO4**	**AN**
3	Find the extreme values of the function $x^3 + 3xy^2 - 3x^2 - 3y^2 + 7$ **[NLJIET]**	**7**	**CO4**	**AN**
4	Find the extreme values for $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$. **(Jan-2019) [NLJIET]**	**7**	**CO4**	**AN**
5	Find local extreme values of the function $f(x,y) = 4x^2 + 9y^2 + 8x - 36y + 24$. **(Jan-2024) [NLJIET]**	**7**	**CO4**	**AN**
6	Find local extreme values of $f(x,y) = xy - x^2 - y^2 - x$. **(Aug-2022) [NLJIET]**	**7**	**CO4**	**AN**
7	Find the local extreme values of the function $f(x,y) = xy - x^2 - y^2 - 2x - 2y + 4$. **(Jun-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
8	Find the extreme values of the function $f(x,y) = x^3 + y^3 + 3x^2 - 3y^2$.**(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
9	Find the maximum and minimum of the function $x^3 + y^3 - 63(x+y) + 12xy$ **[NLJIET]**	**7**	**CO4**	**AN**
10	Find the maxima and minima of the function $f(x,y) = x^2y - xy^2 + 4xy - 4x^2 - 4y^2$ **[NLJIET]**	**7**	**CO4**	**AN**

Topic 8:Method of Lagrange multipliers

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is the necessary condition for the function to have a local extreme? A soldier placed at a point (3, 4) wants to shoot the fighter plane of an enemy which is flying along the curve $y = x^2 + 4$ when it is nearest to him. Find such the distance. **[NLJIET]**	**3**	**CO4**	**AN**

2	Find the point on the plane $x + 2y + 3z = 13$ closest to the point (1,1,1). **[NLJIET]**	**4**	**CO4**	**AN**
3	Find a point on the plane $2x + 3y - z = 5$ which is nearest to the origin, using Lagrange's method of undetermined multipliers. **[NLJIET] OR** Find a point on the plane $2x + 3y - z = 5$ which is nearest to the origin. **[NLJIET]**	**4**	**CO4**	**AN**
4	Find the numbers $x, y, and\ z$ such that $xyz = 8$ and $xy + yz + zx$ is maximum, using the Lagrange's method of undetermined multipliers **[NLJIET]**	**4**	**CO4**	**AN**
5	Use Lagrange's method of multipliers to find the maximum value of $f(x, y, z) = xyz$ if it is given that $x + y + z = 24$ **[NLJIET]**	**4**	**CO4**	**AP**
6	Find the greatest and smallest values that the function $f(x, y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$. **[NLJIET]**	**4**	**CO4**	**AN**
7	Find the shortest distance from the point (1,2,2) to the sphere $x^2 + y^2 + z^2 = 16$. **(Mar-2021) [NLJIET] OR** Use Lagrange method of undetermined multipliers to find the shortest distance from the point (1,2,2) to the sphere $x^2 + y^2 + z^2 = 16$ **[NLJIET]**	**4**	**CO4**	**AN**
8	Find a point within a triangle such that the sum of the squares of its distances from the three vertices is a minimum. **[NLJIET]**	**4**	**CO4**	**AN**
9	Show that the rectangular solid of maximum volume that can be inscribed in a sphere is cube. **[NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3,1,-1)$. **(Jun-2019) [NLJIET]**	**7**	**CO4**	**AN**

2	Find the shortest and longest distance from the point (1,2,-1) to the sphere $x^2+y^2+z^2=24$ **(Jan-2019) [NLJIET]**	**7**	**CO4**	**AN**
3	Find the maximum and minimum distance from the point (1,2,2) to the sphere $x^2+y^2+z^2=36$. **(Jan-2020) [NLJIET]**	**7**	**CO4**	**AN**
4	Determine the minimum value of x^2yz^2 subject to the condition $x+y+2z=5$ using method of Lagrange multipliers. **(Aug-2022) [NLJIET]**	**7**	**CO4**	**AN**
5	Find the minimum value of x^2yz^3 subject to the condition $2x+y+3z=a$ using Lagrange's method of undetermined multipliers. **[NLJIET]**	**7**	**CO4**	**AN**
6	The pressure P at any point (x,y,z) in the space is $\mathrm{P}=400\mathrm{xyz}^2$ .Find the highest pressure at the surface of the unit sphere $\mathrm{x}^2+\mathrm{y}^2+\mathrm{z}^2=1$ **[NLJIET] OR** Suppose that the Celsius temperature at the point (x,y,z) on the sphere $\mathrm{x}^2+\mathrm{y}^2+\mathrm{z}^2=1$is $T=400xyz^2$,Locate the highest and lowest temperature on the sphere. **[NLJIET] OR** The temperature at any point (x,y,z) in space is $T=400xyz^2$. Find the highest temperature on the surface of unit sphere $x^2+y^2+z^2=1$ by the method of Lagrange's multipliers.**[NLJIET]**	**7**	**CO4**	**AN**
7	Find the maximum and minimum values of the function $f(x,y)=3x+4y$ on the circle $\mathrm{x}^2+\mathrm{y}^2=1$ **(Jan-2024) [NLJIET]**	**7**	**CO4**	**AN**
8	Find the extreme values of the function $f(x,y)=x^2+2y^2$ on the circle $\mathrm{x}^2+\mathrm{y}^2=1$.**(Jun-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
9	Find the number $x, yandz$ such that $xyz=16$ and $18xy+16yz+12xz$ is minimum, using the Lagrange's method of undetermined coefficients. **[NLJIET]**	**7**	**CO4**	**AN**

10	You are to construct an open rectangular box from 12 ft^2 of material. What dimensions will result in a box of maximum volume? **[NLJIET] OR** A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box. **[NLJIET]**	**7**	**CO4**	**AN**
11	A rectangular box open at the top is to have a volume of 32 cubic units. Find the dimensions of the box requiring least material for its construction. **[NLJIET] OR** Applying Lagrange's Multipliers to determine dimensions of a rectangular box, open at the top having a volume $32ft^3$ and requiring least amount of material for its construction. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**AN**
12	A rectangular box opens the top is to have a volume of 108 c.c. find the dimension of the box requiring least material for its construction. **[NLJIET]**	**7**	**CO4**	**AN**
13	A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box. **(Jul-2023) [NLJIET]**	**7**	**CO4**	**AN**

MODULE: 5

MULTIVARIABLE CALCULUS (INTEGRATION)

Multiple Integration: Double integrals (Cartesian, Polar), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian, Cylindrical, Spherical).

Topic 1: Evaluation of Double Integrals in Cartesian Coordinates

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level

1	Evaluate $\int_{-1}^{1}\int_{0}^{2}(1-6x^2y)\,dx\,dy$. **[NLJIET]**	3	CO5	EL
2	Evaluate $\int_{1}^{2}\int_{0}^{1}(1+3xy)dxdy$ **[NLJIET]**	3	CO5	EL
3	Evaluate $\int_{0}^{3}\int_{0}^{2}(4-y^2)dydx$ **(Jun-2025-New) [NLJIET]**	3	CO5	EL
4	Evaluate $\int_{0}^{1}\int_{0}^{1}\frac{1}{\sqrt{(1-x^2)(1-y^2)}}dx\,dy$ **[NLJIET]**	3	CO5	EL
5	Evaluate $\int_{y=0}^{1}\int_{x=0}^{2}\frac{1}{\sqrt{4-x^2}\sqrt{1-y^2}}dxdy$. **(Aug-2022) [NLJIET]**	3	CO5	EL
6	Evaluate $\int_{0}^{1}\int_{1}^{2}xy\,dy\,dx$ **(Jan-2020) [NLJIET]**	3	CO5	EL
7	Evaluate $\int_{0}^{1}\int_{1}^{2}\frac{xe^x}{y}\,dydx$ **[NLJIET]**	3	CO5	EL
8	Evaluate $\int_{0}^{2}\int_{0}^{\pi/2}x\,siny\;dydx$.**(Jul-2023) [NLJIET]**	3	CO5	EL
9	Calculate $\iint_R f(x,y)dA$ for $f(x,y)=100-6x^2y$ and $R: 0\le x\le 2, -1\le y\le 1$.**(Jan-2024) [NLJIET]**	3	CO5	EL
10	Evaluate $\iint_R y^2x\,dA$ over the rectangular region $R=\{(x,y): -3\le x\le 2, 0\le y\le 1$.**(Jan-2025-NEW) [NLJIET]**	3	CO5	EL
11	Evaluate $\int_{0}^{3}\int_{0}^{1}(x^2+3y^2)dydx$.**(Jul-2024) [NLJIET]**	3	CO5	EL
12	Evaluate $\int_{0}^{1}\int_{0}^{\sqrt{x}}(x^2+y^2)\,dy\,dx$ **[NLJIET]**	3	CO5	EL
13	Evaluate: $\int_{0}^{1}\int_{0}^{\sqrt{1+x^2}}\frac{dydx}{1+x^2+y^2}$ **[NLJIET]**	3	CO5	EL
14	Evaluate: $\int_{1}^{4}\int_{2x^2}^{3x^2}xe^{x^2+y}\,dy\,dx$ **[NLJIET]**	3	CO5	EL
15	Evaluate $\int_{0}^{1}\int_{0}^{x}e^{\frac{y}{x}}\,dy\,dx$ **[NLJIET]**	3	CO5	EL
16	Evaluate $\int_{0}^{1}\int_{x^2}^{x}(1+xy)\,dy\,dx$ **[NLJIET]**	3	CO5	EL
17	Evaluate $\int_{1}^{ln8}\int_{0}^{lny}e^{x+y}\;dxdy$.**(Jan-2024) [NLJIET]**	3	CO5	EL
18	Evaluate $\int_{0}^{1}\int_{x}^{\sqrt{x}}f(x,y)dydx$ **[NLJIET]**	4	CO5	EL
19	Evaluate $\int_{0}^{1}\int_{0}^{\sqrt{1-y^2}}(x^2+y^2)\,dx\,dy$ **[NLJIET]**	4	CO5	EL
20	Evaluate the integral $\int_{1}^{2}\int_{y}^{2}xy\,dx\,dy$ **[NLJIET]**	4	CO5	EL

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\int_0^3 \int_{\sqrt{\frac{x}{3}}}^1 e^{y^3}\, dydx$ **(Jun-2019) [NLJIET]**	7	CO5	EL

Evaluation of Double Integrals in Polar Coordinates

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\int_0^{\pi} \int_0^{a(1+cos\theta)} rdrd\theta$ **[NLJIET]**	3	CO5	EL
2	Evaluate $\int_0^{\pi} \int_0^{1-cos\theta} r^2 sin\theta\, drd\theta$ **[NLJIET]**	3	CO5	EL
3	Evaluate the integral $\int_0^{\pi/2} \int_0^{1-\sin\theta} r^2 \cos\theta\, drd\theta$ **[NLJIET]**	3	CO5	EL
4	Evaluate $\int_0^{\pi} \int_0^{\sin\theta} rdrd\theta$ **(Jan-2020) [NLJIET]**	3	CO5	EL
5	Evaluate $\int_0^{\pi/2} \int_{a(1-\cos\theta)}^{a} r^2 drd\theta$ **(Mar-2021) [NLJIET]**	4	CO5	EL

Evaluation of Triple Integrals

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\int_0^4 \int_0^4 \int_0^4 (1+xyz)dxdydz$ **[NLJIET]**	3	CO5	EL
2	Evaluate $\int_{-1}^1 \int_0^2 \int_0^1 xz - y^3 dzdydx$ **[NLJIET]**	3	CO5	EL
3	Evaluate the triple integral $\int_0^1 \int_0^{\pi} \int_0^{\pi} y \sin z\, dxdydz.$ **[NLJIET]**	3	CO5	EL
4	Evaluate $\int_0^1 \int_0^{1-y} \int_0^2 dxdydz$ **[NLJIET]**	3	CO5	EL
5	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dxdydz$ **[NLJIET]**	3	CO5	EL
6	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x+y} dzdydx$ **[NLJIET]**	3	CO5	EL
7	Evaluate $\int_0^1 \int_x^1 \int_0^{y-x} dzdydx$ **(Mar-2023) (Jan-2025-New) [NLJIET]**	3	CO5	EL
8	Evaluate $\int_0^1 \int_x^{2x} \int_0^y 2xyz\, dzdydx.$ **(Jul-2023) [NLJIET]**	3	CO5	EL
9	Evaluate $\int_0^1 \int_0^{2-x} \int_0^{2-x-y} dzdydx$ **[NLJIET]**	3	CO5	EL
10	Evaluate $\int_0^1 \int_0^y \int_0^{x+y} ydzdxdy$ **[NLJIET]**	3	CO5	EL

11	Evaluate $\int_0^1 \int_0^x \int_0^{\sqrt{x+y}} z\, dzdydx$ **[NLJIET]**	**3**	**CO5**	**EL**
12	Evaluate $\int_0^1 \int_0^y \int_0^{x+2y} (x+y+z) dzdxdy$ **[NLJIET]**	**3**	**CO5**	**EL**
13	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} dzdydx$ **[NLJIET]**	**3**	**CO5**	**EL**
14	Evaluate $\int_0^1 \int_0^{x^{\frac{1}{4}}} \int_0^{y^2} \sqrt{z} dzdydx$ **(Aug-2022) [NLJIET]**	**3**	**CO5**	**EL**
15	Evaluate the triple integral $\iiint_B xyz^2 dV$ where B is the rectangular box given by $B = \{(x, y, z): 0 \le x \le 1, -1 \le y \le 2, 0 \le z \le 3\}$**(Jun-2025-New) [NLJIET]**	**3**	**CO5**	**EL**
16	Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z\, dzdydx$ **[NLJIET]**	**4**	**CO5**	**EL**
17	Evaluate the integral $\int_0^2 \int_1^2 \int_0^{yz} xyz\, dxdydz$ **(Jul-2024) [NLJIET]**	**4**	**CO5**	**EL**
18	Evaluate the integral $\int_0^2 \int_1^z \int_0^{yz} xyzdxdydz.$ **[NLJIET]**	**4**	**CO5**	**EL**
19	Evaluate: $\int_0^1 \int_0^y \int_0^{x+2y} (x+y+z)^2 dzdxdy$ **[NLJIET]**	**4**	**CO5**	**EL**
20	Evaluate: $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dxdydz$ **[NLJIET]**	**4**	**CO5**	**EL**
21	Evaluate $\int_0^1 \int_0^{1-x} \int_1^{(x+y)^2} xdzdydx$. **(Jun-2019) [NLJIET]**	**4**	**CO5**	**EL**
22	Evaluate $\int_0^e \int_1^{\log y} \int_1^{e^x} \log z\, dzdxdy$ **[NLJIET]**	**4**	**CO5**	**EL**
23	Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dzdydx.$ **(Jan-2018) [NLJIET]**	**4**	**CO5**	**EL**
24	Evaluate $\int_1^e \int_1^{\log y} \int_1^{e^x} (x^2+y^2) dzdxdy$ **(Jan-2019) [NLJIET]**	**4**	**CO5**	**EL**
25	Evaluate the integral $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z}\, dxdydz$ **[NLJIET]**	**4**	**CO5**	**EL**
26	Evaluate $\int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} (r^2 \cos^2\theta + z^2) r d\theta dr dz$ **[NLJIET]**	**4**	**CO5**	**EL**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} e^{-(x^2+y^2+z^2)^{\frac{3}{2}}} dxdydz$ **[NLJIET]**	**7**	**CO5**	**EL**

Topic 2: Evaluation of Double integral over rectangles & general regions

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\iint_R ysin(xy)dA$ where R is the region bounded by $x = 1, x = 2, y = 0\ and\ y = \frac{\pi}{2}$.**(Jan-2019) [NLJIET]**	3	CO5	EL
2	Evaluate $\iint_R (2x - y^2)\, dA$ over the triangular region R enclosed between the lines $y = -x + 1$, $y = x + 1$ & $y = 3$. **[NLJIET]**	3	CO5	EL
3	Evaluate $\iint_S \sqrt{xy - y^2}\, dxdy$Where S is a triangle with vertices (0, 0) ,(10, 1) ,(1, 1). **[NLJIET]**	3	CO5	EL
4	Evaluate $\iint_R e^{2x+3y}\, dA$ where R is triangle bounded by x=0, y=0, x + y =1 **(Mar-2021) [NLJIET]**	4	CO5	EL
5	Integrate $f(x, y) = x^2 + y^2$ over the triangular region with vertices (0,0), (1,0), and (0,1).**[NLJIET]**	4	CO5	EL
6	Evaluate $\iint(x^2 - y^2)dxdy$ over the triangle with vertices $(0,1), (1,1), (1,2)$ **(Jan-2020) [NLJIET]**	4	CO5	EL
7	Sketch the region of integration and evaluate the integral $\iint_R (y - 2x^2)\, dA$, where R is is the region inside the square $\lvert x\rvert + \lvert y\rvert = 1$ **(Jun-2019) [NLJIET]**	4	CO5	EL
8	Sketch the region of integration and evaluate $\int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{\frac{y}{\sqrt{x}}}\, dydx$ **[NLJIET] OR** Evaluate $\int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{\frac{y}{\sqrt{x}}}\, dydx$ **[NLJIET]**	4	CO5	EL
9	Evaluate $\iint_A ydxdy$ where A is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$. **[NLJIET]**	4	CO5	EL
10	Evaluate $\iint_R x\, dA$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$ **[NLJIET]**	4	CO5	EL

11	Evaluate $\iint_R y(x-y^2)dydx$ where R is the region bounded by the curves $y=\sqrt{x}$ and $y=x^3$. **(Mar-2023) [NLJIET]**	**4**	**CO5**	**EL**
12	Evaluate $\iint_R xy\, dA$, where R is the region bounded by x-axis, ordinate $x=2a$ and the curve $x^2=4ay$. **(Aug-2022) [NLJIET]**	**4**	**CO5**	**EL**
13	Evaluate $\iint_{\mathrm{R}}$ xy dA over the area between $y=x^2$ and $y=x$ **[NLJIET]**	**4**	**CO5**	**EL**
14	Evaluate $\iint_R xy(x+y)\, dA$ over the area between $y=x^2$ and $y=x$. **[NLJIET]**	**4**	**CO5**	**EL**
15	Evaluate $\iint (x+y)^2 dxdy$ over the curve bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ **[NLJIET]**	**4**	**CO5**	**EL**
16	Evaluate $\iint y\, dxdy$ over the region bounded by $y=x^2, x=0, x+y=2$ in the first quadrant. **[NLJIET]**	**4**	**CO5**	**EL**
17	Evaluate $\iint_D (x+2y)\, dA$ where D is the region bounded by the parabolas $y=2x^2$ and $y=x+1$ **[NLJIET]**	**4**	**CO5**	**EL**
18	Evaluate $\iint xy\, dxdy$, where the region R is bounded by $x^2+y^2-2x=0, y^2=2x, y=x$.**[NLJIET]**	**4**	**CO5**	**EL**
19	Use double integration to determine the area of the region R inclosed between the parabola $y=\frac{1}{2}x^2$, and the line $y=2x$, also sketch the region. **(Jan-2025-New) [NLJIET]**	**4**	**CO5**	**EL**
20	Evaluate $\iint_R (3x+4y^2)\, dA$ where R is the region in the upper half-plane bounded by the circles $x^2+y^2=1$ and $x^2+y^2=4$.**(Jun-2025-New) [NLJIET]**	**4**	**CO5**	**EL**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\iint_R (x+y)\, dydx$ R is the region bounded by $x=0, x=2, y=x, y=x+2$ **[NLJIET] OR**	**7**	**CO5**	**EL**

	Evaluate the following integral along the region R $\iint_R (x+y)\, dydx$ where R is the region bounded by $x=0, x=2, y=x, y=x+2$. Also, sketch the region. **(Mar-2021) [NLJIET] OR** Sketch the region ***R*** bounded by the lines $x=0, x=2, y=x \;\&\; y=x+2$.Find the area of this region R by double integrals. **(Mar-2023) [NLJIET]**			
2	Evaluate $\iint (x^2+y^2) dxdy$ over the region bounded by the lines $y=4x$, $x+y=3, y=0, y=2$ **[NLJIET]**	**7**	**CO5**	**EL**
3	Calculate $\iint_R \frac{\sin x}{x}\, dA$ Where R is the triangle in the xy-plane bounded by the x-axis, the line $y=x$ and the line $x=1$. **(Jan-2024) [NLJIET]**	**7**	**CO5**	**EL**
4	Evaluate $\iint_R x^2\, dA$, where R is the region in the first quadrant bounded by the hyperbola $xy = 16$ and the lines $y = x, y = 0$ and $x = 8$. **[NLJIET]**	**7**	**CO5**	**EL**

Topic 3: Change of order of integration in double integrals

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Change the order of integration and evaluate $\int_o^a \int_y^a \frac{x}{x^2+y^2} dxdy$ **[NLJIET]**	**4**	**CO5**	**EL**
2	By changing the order of the integration, evaluate $\int_0^3 \int_y^3 \frac{x}{x^2+y^2}\, dxdy$ **(Jan-2019) [NLJIET]**	**4**	**CO5**	**EL**
3	Sketch the region of integration, reverse the order of integration and evaluate the $\int_o^2 \int_0^{4-x^2} \frac{xe^{2y}}{4-y} dydx$ **[NLJIET]**	**4**	**CO5**	**EL**
4	Evaluate $\int_o^\infty \int_x^\infty e^{-y^2} dydx$, by changing the order of integration. Also, show the region of integration given by the limits, clearly. **[NLJIET]**	**4**	**CO5**	**EL**

5	Change the order of Integrations and hence evaluate: $\int_0^1 \int_{2x}^2 e^{y^2} dxdy$. **[NLJIET]**	4	CO5	EL
6	Evaluate the integral $\int_0^2 \int_{\frac{x}{2}}^1 \frac{1}{3} e^{y^2}\ dydx$ by change of order. **(Aug-2022) [NLJIET]**	4	CO5	EL
7	Evaluate the integral $\int_{y=0}^1 \int_{x=0}^{\cos^{-1} y} e^{\sin x}\ dxdy$ by change of order **(Aug-2022) [NLJIET]**	4	CO5	EL
8	Evaluate $\int_0^\infty \int_{-y}^y (y^2 - x^2) e^{-y}\ dxdy$ by changing the order of integration. **(Jan-2019) [NLJIET]**	4	CO5	EL
9	Evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$ by changing the order of integration. **[NLJIET]**	4	CO5	EL
10	Evaluate: $\int_0^2 \int_{y-2}^{2y} xydxdy$ by changing the order of integration. **[NLJIET]**	4	CO5	EL
11	Sketch the region of integration for the integral $\int_0^2 \int_{x^2}^{2x} (4x+2)dydx$ and write an equivalent integral with the order of integration reversed.**(Jan-2024) [NLJIET]**	4	CO5	EL
12	Evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x+y)dxdy$ by changing the order of integration. **[NLJIET]**	4	CO5	EL
13	Evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dydx$ by changing the order of integration. **[NLJIET]**	4	CO5	EL

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate integral $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dydx$ by reversing the order of integration. **[NLJIET] OR** Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dA$ by changing the order of integration. **(Mar-2021) [NLJIET] OR**	7	CO5	EL

	Change the order of integration and hence evaluate the same. Do sketch the region. $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dydx$ **(Mar-2022) [NLJIET]**			
2	Change the order of integration and then evaluate: $\int_0^\infty \int_0^x x\, e^{-\frac{x^2}{y}} dydx$ **[NLJIET]**	**7**	**CO5**	**EL**
3	Apply change of order of integration to evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} dxdy$ also sketch the region**(Jan-2025-New) [NLJIET]**	**7**	**CO5**	**EL**
4	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dydx$. **[NLJIET]**	**7**	**CO5**	**EL**
5	Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x}{\sqrt{1-x^2}\sqrt{1-x^2-y^2}} dxdy$.**(Jul-2024) [NLJIET]**	**7**	**CO5**	**EL**
6	Change the order of integration and hence evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xydydx$ **[NLJIET]**	**7**	**CO5**	**EL**
7	Change the order of integration and evaluate $\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dydx$ **[NLJIET]**	**7**	**CO5**	**EL**
8	Evaluate: $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dydx$ By changing the order of integration. **[NLJIET]**	**7**	**CO5**	**EL**
9	Evaluate the integral $\int_0^1 \int_{1-\sqrt{1-y^2}}^{1+\sqrt{1-y^2}} dxdy$ by the changing the order of integration, Give the sketch of regions of integration. **(Mar-2023) [NLJIET]**	**7**	**CO5**	**EL**
10	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^1 \int_x^1 \sin(y^2) dydx$ **(Jan-2020)(Jun-2025-New) [NLJIET]**	**7**	**CO5**	**EL**
11	Sketch the region of integration, change the order of integration and evaluate $\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3+1} dydx$. **(Jul-2023) [NLJIET]**	**7**	**CO5**	**EL**

12	Change the order of integration and then evaluate: $\int_0^{\frac{a}{\sqrt{2}}}\int_x^{\sqrt{a^2-x^2}} y^2 dA$ **[NLJIET]**	7	CO5	EL
13	Change the order of integration and then evaluate $\int_0^1\int_{x^2}^{2-x} xy\ dydx$ **[NLJIET]**	7	CO5	EL

Topic 4: Change of variables (Cartesian to polar)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate the integral $I=\int_0^1\int_0^x \sqrt{x^2+y^2}\ dydx$ By passing onto the polar coordinates. **[NLJIET]**	3	CO5	EL
2	Evaluate: $\int_0^1\int_0^1 dxdy$ by changing to polar co-ordinates. **[NLJIET]**	3	CO5	EL
3	Evaluate $\iint_R (x^2+y^2)xdxdy$ over the positive quadrant of the circle $x^2+y^2=a^2$ by polar coordinates. **[NLJIET]**	3	CO5	EL
4	Use polar coordinates to evaluate $\iint_R (x+y)dxdy$ over the positive quadrant of the circle $x^2+y^2=a^2$ **[NLJIET]**	3	CO5	EL
5	Evaluate $\iint_R xy\sqrt{x^2+y^2}dxdy$ by changing into polar coordinates where $R=\{(x,y): 1\le x^2+y^2\le 4, x\ge 0, y\ge 0\}$ **(Mar-2023) [NLJIET]**	3	CO5	EL
6	Evaluate $\int_0^2\int_0^{\sqrt{4-x^2}}(x^2+y^2)dydx$ by changing into the polar coordinates. **(Jan-2019) [NLJIET]**	4	CO5	EL
7	Evaluate $\int_0^a\int_0^{\sqrt{a^2-x^2}}(x^2+y^2)\ dydx$ by changing into polar coordinates where a > 0 **[NLJIET]**	4	CO5	EL
8	Change in to polar coordinates then solve $\int_0^2\int_0^{\sqrt{4-x^2}} e^{-(x^2+y^2)}\ dydx$.**(Aug-2022) [NLJIET]**	4	CO5	EL
9	Change into polar co-ordinate and evaluate $\int_0^a\int_0^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)}\ dydx$ **[NLJIET]**	4	CO5	EL

10	Evaluate the integral $\int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}dxdy$ by changing into polar coordinates **[NLJIET] OR** By changing to polar coordinates, evaluate $\int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}dxdy$. Also sketch the regions. **[NLJIET]**	**4**	**CO5**	**EL**
11	Evaluate the following integral by changing into polar coordinates. $\iint_R e^{x^2+y^2}dA$ where R is the semi-circular region bounded by the x-axis and the cdurve $y=\sqrt{1-x^2}$. **(Jan-2025-New) [NLJIET]**	**4**	**CO5**	**EL**
12	Change into polar coordinates and hence evaluate: $\int_0^a\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} y^2dydx$ **[NLJIET]**	**4**	**CO5**	**EL**
13	Sketch the region of integration and change in to polar integral and then evaluate. $\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dydx$ **[NLJIET]**	**4**	**CO5**	**EL**
14	Evaluate the integral $\int_0^{4a}\int_{\frac{y^2}{4a}}^{y}\frac{x^2-y^2}{x^2+y^2}dxdy$by transforming into Polar coordinates **[NLJIET]**	**4**	**CO5**	**EL**
15	Evaluate $\int_0^a\int_y^a\frac{x}{x^2+y^2}dxdy$ by transforming into polar coordinates. **[NLJIET]**	**4**	**CO5**	**EL**
16	Evaluate the integrals$\int_0^{\infty}\int_0^{\infty}\frac{1}{(1+x^2+y^2)^2}dxdy$ **(Jan-2019) [NLJIET]**	**4**	**CO5**	**EL**
17	Evaluate$\int_0^a\int_0^{\sqrt{a^2-y^2}} y^2\sqrt{x^2+y^2}\,dydx$, by changing into polar coordinates. **[NLJIET]**	**4**	**CO5**	**EL**
18	Integrate$f(x,y)=\frac{\ln(x^2+y^2)}{\sqrt{x^2+y^2}}$ over the region $1\le x^2+y^2\le e$; by changing to polar coordinates. **[NLJIET]**	**4**	**CO5**	**EL**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}(x^2+y^2)dydx$ by changing into the polar coordinates.**(Jan-2024) [NLJIET]**	**7**	**CO5**	**EL**

2	Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{x^2+y^2}\, dydx$ by changing into polar coordinates.**(Jul-2024) [NLJIET]**	7	CO5	EL

Topic 5: Applications: areas

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
	Area Using Cartesian Coordinates			
1	Find the area of the circle $x^2 + y^2 = 1$ using double integration. **(Dec-2013) [NLJIET]**	3	CO5	AN
2	Find the area of the region enclosed by $y = x^2$, $y = 2x - x^2$ **(Jan-2019) [NLJIET]**	3	CO5	AN
3	Show by double integration that area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. **[NLJIET]**	4	CO5	AN
4	Find the area of the region R bounded by $y = 2x^2$ and $y^2 = 4x$ **(Jun-2025-New) [NLJIET]**	4	CO5	AN
5	Using double integration find area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ **[NLJIET]**	4	CO5	AN
6	Find the area of circle $x^2 + y^2 = a^2$ by double integration. **[NLJIET]**	4	CO5	AN
7	Find the area of the region R bounded $y = x \text{ and } y = x^2$ in the first quadrant. **(Jan-2024) [NLJIET]**	4	CO5	AN
8	Find the area of the region covered by $x = 1, x = 4, y = 0$ and $y = \sqrt{x}$. **(Aug-2022) [NLJIET]**	4	CO5	AN
9	Find the area of the region bounded by the curves $y = sinx, y = cosx$ And the lines $x = 0$ and $x = \frac{\pi}{4}$. **(Jun-2019) [NLJIET]**	4	CO5	AN
10	Find the area of region R enclosed by the line $y = x, x = 1$ and the x-axis.**[NLJIET]**	4	CO5	AN

11	Find the area of the region R enclosed by the parabola $y = x^2$ and the line $y = x + 2$ **[NLJIET]**	**4**	**CO5**	**AN**
12	Find area enclosed within the curves $y = 2 - x \text{ and } y^2 = 2(2 - x)$ **(Mar-2021) [NLJIET]**	**4**	**CO5**	**AN**
	Area Using Polar Coordinates			
1	Evaluate $\iint_R r^3 \sin 2\theta dr d\theta$ over the area bounded in the first quadrant between the circles r = 2 and r = 4. **[NLJIET]**	**3**	**CO5**	**EL**
2	Find the total area enclosed by lemniscate $r^2 = a^2 \cos 2\theta$ **[NLJIET]**	**3**	**CO5**	**AN**
3	Evaluate $\iint r^3 dr d\theta$ over the area bounded between the circles $r = 2cos\theta$ and $r = 4cos\theta$ **[NLJIET]**	**3**	**CO5**	**EL**
4	Evaluate integral $\iint_R \frac{r dr d\theta}{\sqrt{a^2+r^2}}$ over the region R which is one loop of $r^2 = a^2 \cos 2\theta$ **(Jan-2019) [NLJIET]**	**3**	**CO5**	**EL**
5	Using double integrations, find the area enclosed by the cardioid $r = a(1 + cos\theta)$.**[NLJIET]**	**3**	**CO5**	**AN**
6	Evaluate the area between the circles $r = 2\cos\theta$ and $r = 4\cos\theta$, using double integration. **[NLJIET]**	**3**	**CO5**	**EL**
7	Find the area lying inside the circle $r = a \sin \theta$ and outside the cardioids $r = a(1 - cos\ \theta)$. **[NLJIET]**	**3**	**CO5**	**AN**
8	Find the area common to the cardioid $r = a(1 - cos\theta)$ and $r = a(1 + cos\theta)$. **[NLJIET]**	**4**	**CO5**	**AN**
9	Find the area enclosed by the cardioid $r = 1 + cos\theta$. **[NLJIET]**	**4**	**CO5**	**AN**
10	Find the area lying inside the cardioid $r = a(1 + cos\theta)$ and outside the region of the circle $r = a$, by double integration. **[NLJIET]**	**4**	**CO5**	**AN**
11	Evaluate $\iint_R sin\theta dA$, where R is the region in the first quadrant that is outside the circle $r = 2$ and inside the cardioids $r = 2(1 + cos\theta)$**[NLJIET]**	**4**	**CO5**	**EL**

12	Find the area outside the circle $r = 2a\,cos\theta$ and inside the cardioid $r = a(1 + cos\theta)$. **[NLJIET]**	**4**	**CO5**	**AN**
13	Find the area common to the circles r = a and circle $r = 2a\,cos\theta$ using double integration.**[NLJIET]**	**4**	**CO5**	**AN**
14	Find the area of the region that lies inside the cardioid $r = 1 + cos\theta$ and outside the circle $r = 1$. **[NLJIET]**	**4**	**CO5**	**AN**
15	Find the area common to both of the circles $r = cos\theta \text{ and } r = sin\theta$ **[NLJIET]**	**4**	**CO5**	**AN**
16	Evaluate $\iint rsin\theta drd\theta$ over the area bounded by the cardioid $r = a(1 + cos\theta)$. above the initial line. **[NLJIET]**	**4**	**CO5**	**EL**

Topic 6: Applications: volumes

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Use triple integral to find the volume of the cylinder $x^2 + y^2 = 1$ between the places $z = 1 \text{ and } z = 2$. **[NLJIET]**	**3**	**CO5**	**AN**
2	Find the volume of the tetrahedron bounded by the plane $x + y + z = 2$ and the planes $x = 0, y = 0 \text{ and } z = 0$. **[NLJIET]**	**3**	**CO5**	**AN**
3	Find the volume bounded by cylinder $x^2 + y^2 = 4$ and the plane $y + z = 3$ and $z = 0$. **[NLJIET]**	**4**	**CO5**	**AN**
4	Find by double integrations, the volume of the cylinder $x^2 + y^2 = 1$ between the planes $z = 0$ and $y + z = 2$ **[NLJIET]**	**4**	**CO5**	**AN**
5	Find the volume by triple integrals of the solid S that is bounded by the plane $2x + 3y + 4z = 12$ and the three coordinate planes. **[NLJIET]**	**4**	**CO5**	**AN**
6	Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 3$ and z = 0 **[NLJIET]**	**4**	**CO5**	**AN**
7	Find the volume bounded by the cylinder $x^2 + z^2 = 1, y = 0$ and $y + z = 3$ using triple integration. **[NLJIET]**	**4**	**CO5**	**AN**

8	Evaluate $\iiint z(x^2+y^2)dv$ over the volume of the cylinder $x^2+y^2=1$ intercepted by the planes $z = 2$ and $z = 3$. **[NLJIET]**	**4**	**CO5**	**EL**
9	Find the volume of the region that lies under the paraboloid $z=x^2+y^2$ and above the triangle enclosed by the lines $y=x, x=0$ and $x+y=2$ in the xy- plane. **[NLJIET]**	**4**	**CO5**	**AN**
10	Find the volume of the region bounded by the surface $x=0, y=0, x+y+z=1$ and $z=0$. **[NLJIET]**	**4**	**CO5**	**AN**
11	Find the volume of the region D enclosed by surfaces $z=x^2+3y^2$ and $z=8-x^2-y^2$ **(Jan-2019) [NLJIET]**	**4**	**CO5**	**AN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Evaluate $\iiint_E 2x\,dV$ where E is the region under the plane $2x+3y+z=6$ that lies in the first octant. **[NLJIET]**	**7**	**CO5**	**EL**
2	Evaluate $\iiint xyz\,dxdydz$ over the positive octant of the sphere $x^2+y^2+z^2=4$ **(Jan-2020) [NLJIET]**	**7**	**CO5**	**EL**
3	Evaluate $\iiint z^2dxdydz$ over the region common to the sphere $x^2+y^2+z^2=4$ and the cylinder $x^2+y^2=2x$ **[NLJIET]**	**7**	**CO5**	**EL**
4	Find the volume under the plane $x+y+z=6$ and above the triangle in the xy- plane bounded. **(Dec-2012) [NLJIET]**	**7**	**CO5**	**AN**
5	Use triple integral to find the volume of the solid within the cylinder $x^2+y^2=9$ between the planes $z=1and$ $x+z=1$. **[NLJIET]**	**7**	**CO5**	**AN**
6	Find the volume below the surface $z=x^2+y^2$, above the plane $z=0$ and inside the cylinder $x^2+y^2=2y$. **(Jan-2019) [NLJIET]**	**7**	**CO5**	**AN**
7	Find the volume of the sphere $x^2+y^2+z^2=a^2$ **[NLJIET]**	**7**	**CO5**	**AN**
8	Find the volume of the solid that lies under the paraboloid $z=x^2+y^2$ and above the region R in the xy-plane bounded by the	**7**	**CO5**	**AN**

	lines $y = 2x$ and the parabola $y = x^2$.**(Jun-2025-New)** **[NLJIET]**			

Topic 7: Applications: Center of mass and Gravity (constant and variable densities)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the center of mass of a thin uniform plate whose shape is the region between $y = cosx$ and the $x - axis$ $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$. Take the density is constant,$\sigma(x, y) = 1$.**[NLJIET]**	4	CO5	AN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Find the center of gravity of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant whose density is $kxyz$.**[NLJIET]**	7	CO5	AN
2	Find the center of mass of two dimensional plate that occupies the triangle $0 \le x \le 1$, $0 \le y \le x$, and has density function xy. **[NLJIET]**	7	CO5	AN
3	A thin plate covers the triangular region bounded by the x-axis and the line $x = 1$ and $y = 2x$ in the first quadrant.The plane density at the point (x, y) is $\delta(x, y) = 6x + 6y + 6$ find the plate's mass, first moments and the centre of mass about the coordinate axis.**(Jan-2025-New) [NLJIET]**	7	CO5	AN

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